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Properties of the Z^0 Boson

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M766

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Abstract

Between 1989 and 2000, the Large Electron-Positron Collider (LEP) at CERN became a hub for global scientific efforts to study electron-positron collisions. These experiments, conducted at unprecedented energy levels, marked the first successful observation of Z^0 bosons, solidifying the foundation of the Standard Model and its reliance on gauge theory. The Standard Model explains particle interactions through quantum field theories governed by local gauge symmetries, with coupling constants defining the interaction strengths. Due to its minimal effect at the particle scale, gravity is disregarded in these studies. The electroweak theory, proposed by Glashow, Salam, and Weinberg, gained experimental confirmation in 1983 with the discovery of the $W^\pm$ and Z^0 bosons at CERN. In contrast, the strong force is described by Quantum Chromodynamics (QCD), which involves interactions mediated by eight massless gluons. This study seeks to examine the Z^0 boson's characteristics and decay patterns using experimental data from OPAL.

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1 Properties of the Z^0 -Boson

1.1 Role of the Z^0 Boson within the Standard Model

The Z^0 -boson is a fundamental mediator of the weak force, playing a central role in the Standard Model. It facilitates neutral current interactions, where particles exchange a Z^0 -boson without altering their electric charge. This mechanism differs from charged current interactions mediated by $W^\pm$ bosons. Alongside the photon, $W^\pm$ bosons, and the Higgs boson, the Z^0 -boson forms the electroweak interaction framework governed by the $SU(2) \times U(1)$ gauge symmetry. Its mass, approximately 91.2 GeV, is acquired through spontaneous symmetry breaking via the Higgs mechanism.

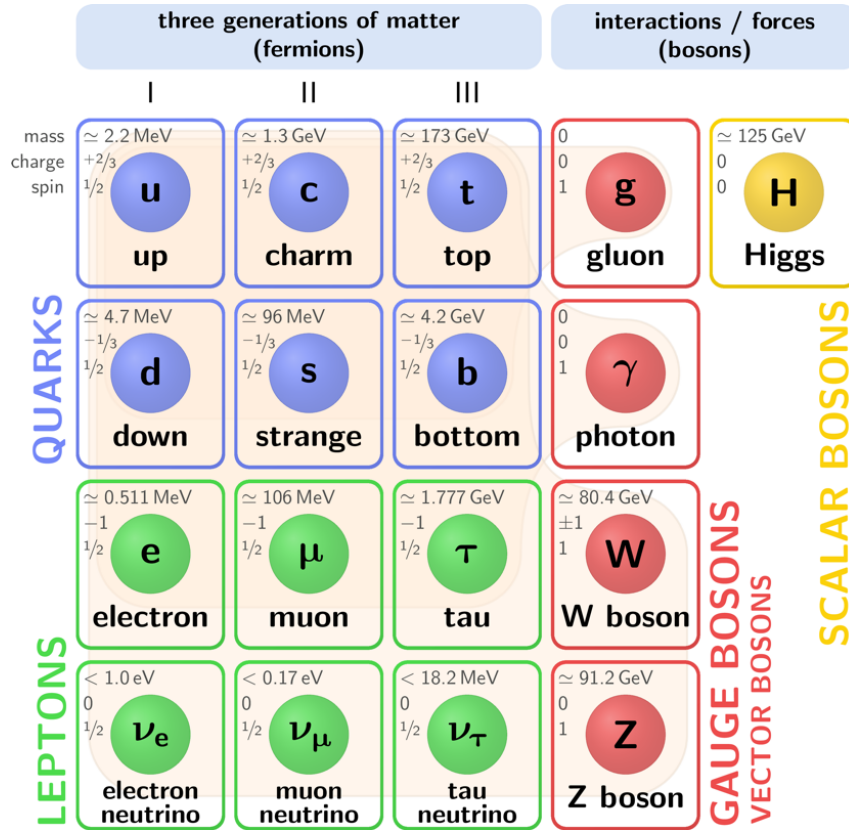


Figure 1.1: The Standard Model of particle physics, displaying the three generations of matter (quarks and leptons) alongside the force-mediating bosons and the Higgs boson.

The Z^0 -boson's properties and behavior confirm the electroweak unification described by Glashow, Salam, and Weinberg. It is unique in coupling to all fermions, including neutrinos, a feature the photon lacks. This universality has made the Z^0 -boson pivotal for verifying Standard Model predictions.

1.2 Particle Couplings of the Z^0 Boson

The Z^0 -boson is a neutral gauge boson that plays a critical role in the electroweak unification described by the $SU(2) \times U(1)$ symmetry. It emerges as a quantum-mechanical mixture of the W_0 and Y_0 bosons, leading to the formation of the Z^0 -boson and the photon (γ) through field mixing. This mixing is governed by the Weinberg angle (θ_W):

$$|\gamma\rangle = \cos \theta_W |Y^0\rangle + \sin \theta_W |W^0\rangle \quad (1.1)$$

$$|Z^0\rangle = -\sin \theta_W |Y^0\rangle + \cos \theta_W |W^0\rangle \quad (1.2)$$

Although the Z^0 -boson and the photon originate from the same base structure and are described by an $SU(2)$ algebra, their properties differ significantly. The photon is massless and does not couple to neutrinos, whereas the Z^0 -boson has a mass of approximately $91 \text{ GeV}/c^2$ and can couple to neutrinos, making it unique among the gauge bosons.

(a) Fermions							
	Fermionmultipletts			I	I_3	Q_f	color
Leptons	$\begin{pmatrix} \nu_e \\ e \\ e_R \end{pmatrix}_L$	$\begin{pmatrix} \nu_\mu \\ \mu \\ \mu_R \end{pmatrix}_L$	$\begin{pmatrix} \nu_\tau \\ \tau \\ \tau_R \end{pmatrix}_L$	$1/2$	$+1/2$ $-1/2$ 0	0 -1 -1	$—$ $—$
	$\begin{pmatrix} u \\ d' \\ u_R \\ d_R \end{pmatrix}_L$	$\begin{pmatrix} c \\ s' \\ c_R \\ s_R \end{pmatrix}_L$	$\begin{pmatrix} t \\ b' \\ t_R \\ b_R \end{pmatrix}_L$	$1/2$	$+1/2$ $-1/2$ 0 0	$+2/3$ $-1/3$ $+2/3$ $-1/3$	r, g, b

(b) Bosons as transmitters of interaction

Interacton	Charge	Mediator	Mass [GeV/c ²]
strong	color charge	Gluon	0
electromagnetic	electr. charge	Photon γ	0
weak	weak charge	$W^\pm, Z^0$	80 and 91
gravity	mass	Graviton	0

Figure 1.2: (a) Fermion multiplets displaying leptons and quarks with their respective properties, and (b) bosons as mediators of interactions with their charges and masses.

The relationship between the masses of the W and Z^0 bosons is described by the weak mixing angle:

$$M_W^2 \sin^2 \theta_W = \frac{\pi \alpha}{\sqrt{2} G_F} \quad (1.3)$$

where α is the fine structure constant, and G_F is the Fermi coupling constant. Furthermore, the masses of the W and Z^0 bosons are related by:

$$\frac{M_W}{M_Z} = \cos \theta_W \quad (1.4)$$

1.3 Coupling Mechanism

The Z^0 -boson couples universally to all fermions, with the coupling strength determined by the fermion's weak isospin (I_3) and their electric charge (Q). These couplings depend on two types of interactions:

- **Vector Coupling (g_V):** Associated with the symmetric part of the fermion's wavefunction.
- **Axial-Vector Coupling (g_A):** Related to the asymmetric part.

The vector and axial-vector couplings for a fermion f are defined as:

$$g_V^f = I_3^f - 2Q^f \sin^2 \theta_W, \quad g_A^f = I_3^f \quad (1.5)$$

where $\sin^2 \theta_W$ is the weak mixing angle, a parameter that quantifies the mixing of the weak and electromagnetic forces in the electroweak theory.

Neutrinos: As neutral particles, neutrinos only couple through the axial term since their electric charge (Q) is zero.

Charged Fermions: Both vector and axial-vector couplings contribute to their interactions with the Z^0 -boson.

The coupling constants are pivotal in determining the partial decay widths of the Z^0 -boson into various fermionic final states, as they dictate the probabilities of such decays.

1.4 Production in High Energy Collisions

The Z^0 -boson can be generated at particle accelerators through processes like fermion-antifermion annihilation. A particularly effective mechanism is electron-positron annihilation:

$$e^- + e^+ \rightarrow (\gamma, Z^0) \rightarrow f \bar{f} \quad (1.6)$$

where e^- and e^+ represent the electron and positron, and $f \bar{f}$ denotes the produced fermion-antifermion pair. The reaction occurs via the exchange of a neutral gauge boson (γ or Z^0), with the creation of a Z^0 -boson requiring the center-of-mass energy to match or exceed its rest mass ($M_{Z^0} = 91.188 \text{ GeV}/c^2$).

This energy dependence makes Z^0 -production highly efficient in accelerators tuned to the Z^0 -resonance, as seen in the LEP collider experiments. The sharp peak in production probability at the resonance energy is a hallmark of the Z^0 -boson's unique coupling properties.

The ability of the Z^0 -boson to couple to all fermions, including neutrinos, is central to its role in testing the Standard Model. The coupling constants directly influence the partial decay widths of the Z^0 -boson, determining the branching ratios into leptonic, hadronic, and invisible decay channels.

1.5 Decay Properties of the Z^0 Boson

The Z^0 -boson is unstable and decays rapidly into fermion-antifermion pairs via weak neutral currents. The possible decay channels are classified into three major types:

1. Leptonic Decays

These occur when the Z^0 -boson decays into pairs of charged leptons (e, μ, τ) or neutrinos (ν_e, ν_μ, ν_τ):

$$Z^0 \rightarrow e^+ e^-, \mu^+ \mu^-, \tau^+ \tau^-$$

$$Z^0 \rightarrow \nu_e \bar{\nu}_e, \nu_\mu \bar{\nu}_\mu, \nu_\tau \bar{\nu}_\tau$$

Branching Ratios:

- Each charged lepton channel accounts for approximately 3.4% of the total decay width.
- Invisible decays into neutrino-antineutrino pairs contribute about 20% to the total width, making them a significant yet indirect signature of Z^0 -decay.

2. Hadronic Decays

These dominate Z^0 -boson decays and involve quark-antiquark pairs:

$$Z^0 \rightarrow q\bar{q} \quad (q = u, d, c, s, b)$$

- The total hadronic branching ratio is approximately 70%, with individual contributions varying among quark flavors.
- Top quarks (t) are excluded because their mass ($m_t \approx 172 \text{ GeV}$) exceeds the Z^0 mass.

The hadronic decays result in jets due to the subsequent fragmentation of quarks into hadrons. The gluon emissions during fragmentation can also generate multi-jet events.

3. Invisible Decays

Invisible decays involve neutrino-antineutrino pairs ($Z^0 \rightarrow \nu\bar{\nu}$). Although these particles do not interact directly with the detector, their presence is inferred from missing energy and momentum in the final state.

1.5.1 Decay Widths

The decay width (Γ) is a critical parameter for understanding the Z^0 -boson's lifetime and branching ratios. It represents the sum of partial decay widths (Γ_f) across all possible final states:

$$\Gamma_Z = \Gamma_e + \Gamma_\mu + \Gamma_\tau + \Gamma_\nu + \Gamma_{\text{had}} + N_\nu \Gamma_\nu + \Gamma_{\text{unknown}} \quad (1.7)$$

where Γ_e , Γ_μ , and Γ_τ correspond to the decay channels of the electron, muon, and tau, respectively, Γ_{had} encompasses all quark decay contributions, N_ν represents the neutrino-antineutrino pairs, and Γ_{unknown} accounts for any decay processes not predicted by the Standard Model.

The partial decay widths are proportional to the square of the couplings:

$$\Gamma_f \propto (g_V^f)^2 + (g_A^f)^2 \quad (1.8)$$

Key Widths:

- **Leptonic Widths (Γ_l):** Each charged lepton channel contributes approximately 83 MeV.
- **Neutrino Widths (Γ_ν):** Each neutrino flavor contributes approximately 167 MeV.
- **Hadronic Widths (Γ_{had}):** Hadronic decays contribute approximately 1.74 GeV.

The total decay width of the Z^0 -boson is approximately 2.5 GeV.

1.5.2 Branching Ratios

The branching ratios (BR) express the fraction of decays into specific final states. For the Z^0 -boson:

- **Hadronic decays** dominate with $\text{BR}_{\text{had}} \approx 70\%$
- **Leptonic decays** contribute $\text{BR}_l \approx 10\%$ collectively for charged leptons
- **Invisible decays** contribute $\text{BR}_{\text{inv}} \approx 20\%$

These ratios reflect the relative coupling strengths and are consistent with the Standard Model predictions.

1.6 Z^0 -boson Resonance and Breit–Wigner Distribution

The likelihood of creating a Z^0 -boson or photon during collisions is described by the cross section:

$$\sigma_{e^+e^- \rightarrow f\bar{f}} = \sigma_Z + \sigma_\gamma + \sigma_{Z\gamma} \quad (1.9)$$

At the resonance energy of the Z^0 -boson, photon production is largely suppressed, making Z^0 -boson generation dominant. The distribution governing Z^0 -boson exchange, known as the Breit-Wigner distribution, describes the cross section for producing a fermion-antifermion pair ($f\bar{f}$) through Z^0 exchange:

$$\sigma_f = \frac{12\pi\Gamma_e\Gamma_f}{sM_Z^2} \cdot |\chi(s)|^2$$

or equivalently:

$$\sigma_f = \frac{12\pi}{M_Z^2} \cdot \frac{s\Gamma_e\Gamma_f}{(s - M_Z^2)^2 + \left(\frac{s\Gamma_Z}{M_Z}\right)^2} \quad (1.10)$$

Here:

- $s = E_{\text{CM}}^2$ represents the squared center-of-mass energy,
- $M_Z = 91.188 \text{ GeV}/c^2$ is the Z^0 -boson mass,
- Γ_e and Γ_f correspond to the partial widths related to the Z^0 -boson coupling with the electron-positron pair and the final fermion-antifermion pair, respectively,
- $\chi(s)$ is the propagator.

At the resonance energy ($s = M_Z^2$), the cross section simplifies to:

$$\sigma_{\text{peak},f} = \frac{12\pi}{M_Z^2} \cdot \frac{\Gamma_e \Gamma_f}{\Gamma_Z^2} \quad (1.11)$$

Figure 1.3 shows cross sections near the Z^0 -boson resonance. The solid line represents the predictions from the Standard Model, while the data points reflect experimental results from various accelerators.

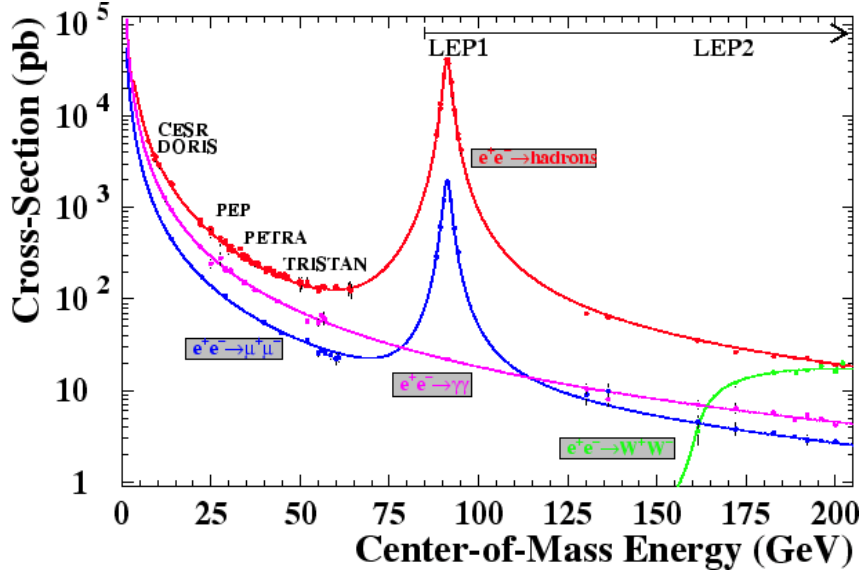


Figure 1.3: Cross sections near the Z^0 -boson resonance. The solid line represents the predictions from the Standard Model, while the data points reflect experimental results from various accelerators.

By fitting the Breit-Wigner curve to the measured cross section σ_f as a function of the center-of-mass energy (E_{CM}) near the resonance, the following Z^0 -boson properties can be extracted:

- **Mass (M_Z):** Corresponds to the position of the peak.
- **Total Decay Width (Γ_Z):** Indicates the range of energies over which the Z^0 -boson can be produced, with a broader width implying a shorter lifetime.
- **Peak Cross Section ($\sigma_{\text{peak},f}$):** Represents the maximum height of the cross section.

2 Detection and Identification of Particles

The detection and identification of particles in high-energy physics experiments involve observing the by-products of particle interactions, as the particles themselves are often too short-lived to be directly measured. This process relies on advanced detector technologies and precise measurements of the particles' properties, such as energy, momentum, charge, and trajectories.

2.1 The OPAL Detector and its Sub-Systems

The OPAL (Omni-Purpose Apparatus at LEP) detector, located at CERN's Large Electron-Positron Collider (LEP), was a versatile instrument designed to study a wide array of particle physics phenomena. Its primary purpose was to analyze the outcomes of electron-positron collisions, providing insights into fundamental interactions and particle properties.

The LEP was capable of accelerating particles to a center-of-mass energy of approximately $E_{\text{CM}} = \sqrt{s} \approx 110 \text{ GeV}/c^2$, which enabled the production of Z^0 -bosons with a mass of $M_{Z^0} = 91.188 \text{ GeV}/c^2$. Following an upgrade, the LEP achieved $E_{\text{CM}} = \sqrt{s} \approx 200 \text{ GeV}/c^2$, allowing for the production of $W^\pm$ -bosons as well.

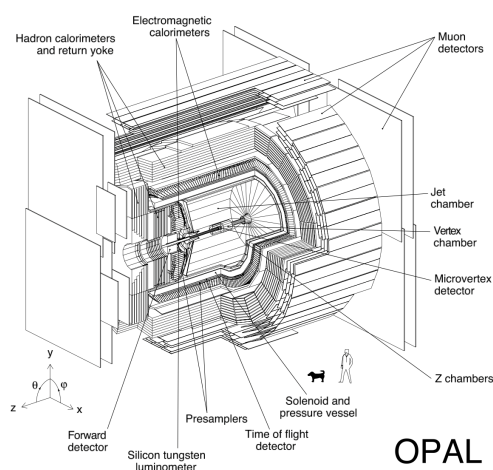


Figure 2.1: Schematic diagram of the OPAL detector.

2.1.1 Tracking Detector

The tracking detector in the OPAL setup is designed to record the paths of charged particles as they traverse through its structure. This is achieved by detecting ionization events caused by charged particles interacting with the detector material. As the particles pass through, they ionize atoms along their path, generating electron-ion pairs. These ionization events are captured by the detector, allowing the reconstruction of the particles' trajectories. By analyzing this data, the momentum and electric charge of the particles can be accurately determined.

2.1.2 Electromagnetic Calorimeter

The electromagnetic calorimeter is specifically engineered to measure the energy of particles interacting via electromagnetic forces, such as photons and electrons. It absorbs the energy of incoming particles, initiating a cascade of secondary particles known as electromagnetic showers. The calorimeter detects these showers using techniques like scintillation or Cherenkov radiation and calculates the total energy deposited. This measurement is vital for identifying and quantifying the energy of photons and electrons.

2.1.3 Hadronic Calorimeter

The hadronic calorimeter focuses on measuring the energy of particles that interact through the strong nuclear force, such as protons, neutrons, and pions. When hadrons enter the calorimeter, their energy is absorbed, resulting in hadronic showers—a series of secondary particles generated by the interactions. The calorimeter uses dense absorber materials to enhance the likelihood of interaction, capturing the total energy deposited by the hadrons. This information helps determine the energy and type of hadronic particles.

2.1.4 Muon Chambers

Muon chambers are specialized components of the OPAL detector designed to identify muons—charged leptons known for their ability to penetrate deeper into matter than most other particles due to their minimal interactions. These chambers are typically positioned outside the calorimeters and consist of layers of tracking materials. As muons pass through these layers, they ionize the detector material, and the resulting signals are recorded to trace their trajectories. This tracking capability is essential for distinguishing muons from other particle types and measuring their energy.

2.2 Particle Detection Techniques and Z^0 -Boson Decay Channels

The OPAL detector employs its various subsystems to identify different types of particles and analyze the decay channels of the Z^0 -boson. Each particle type interacts with the detector components in distinct ways, leaving unique signatures that enable their identification.

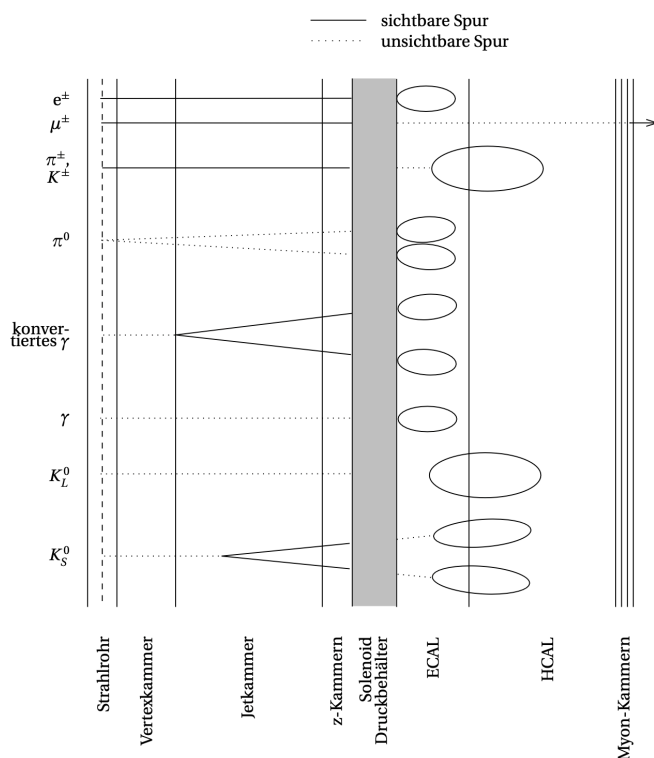


Figure 2.2: Schematic representation of particle detection techniques within the OPAL detector, illustrating the interaction of various particle types with the detector subsystems, including the tracking chambers, electromagnetic calorimeters (ECAL), hadronic calorimeters (HCAL), and muon chambers.

2.2.1 Charged and Neutral Particles

Charged particles, such as electrons, muons, and charged hadrons, interact with the tracking detector by ionizing its material. This ionization creates measurable tracks that can be analyzed to determine the particles' trajectories. In contrast, neutral particles like photons and neutrons do not leave tracks in the tracking detector:

- **Photons:** Detected in the electromagnetic calorimeter by producing electromagnetic showers.
- **Neutrons:** Identified indirectly through their interactions in the hadronic calorimeter.

These differences in interaction behavior help distinguish between charged and neutral particles during data analysis.

2.2.2 Hadrons vs. Leptons

Hadrons, such as protons and pions, interact strongly with the hadronic calorimeter, creating extensive hadronic showers. Leptons, on the other hand, primarily interact via the electromagnetic or weak forces:

- **Electrons:** As leptons, they generate electromagnetic showers in the electromagnetic calorimeter.
- **Muons:** Deposit minimal energy in the calorimeters but penetrate deeply, eventually reaching the muon chambers, where they are detected by ionization events.

This differentiation between interaction types allows precise identification of hadrons and leptons.

2.2.3 Challenges in Particle Detection

Some particles in the Z^0 -boson decay channels are difficult to detect directly. Neutrinos, for example, interact only via the weak force, making them incapable of ionizing detector material or producing observable showers. As a result, neutrinos are "invisible" to the detector, and their presence can only be inferred indirectly by analyzing missing energy and momentum in the event.

2.2.4 Decay Channels of the Z^0 -Boson

The Z^0 -boson decays into various final states, including lepton pairs, hadron pairs, and neutrino pairs. Each decay channel leaves distinct patterns in the detector subsystems:

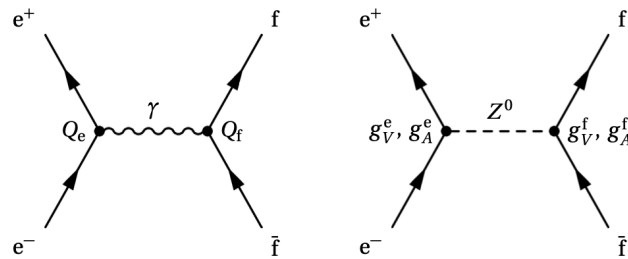


Figure 2.3: Feynman diagrams for the decay of the Z^0 -boson into fermion-antifermion pairs through photon (γ) and Z^0 -boson exchange.

- **Lepton Pairs:** Decays into electron-positron or muon-antimuon pairs are identified by analyzing the characteristic tracks in the tracking detector, along with energy deposits in the electromagnetic calorimeter or signals in the muon chambers.
- **Hadron Pairs:** Hadronic decays result in substantial energy deposits in the hadronic calorimeter and multiple tracks in the tracking detector, corresponding to the fragmentation of quarks into jets of hadrons.
- **Invisible Decays:** Neutrino-antineutrino pairs are detected indirectly by identifying missing energy and momentum, as the neutrinos escape without leaving detectable signals.

By combining data from the different subsystems, the OPAL detector is able to reconstruct and classify the decay channels of the Z^0 -boson with high accuracy, furthering the understanding of particle interactions and validating the predictions of the Standard Model.

3 Variables and Analysis

3.1 Cross Section

In particle physics, the cross section (σ) provides a quantitative measure of the likelihood of a specific interaction occurring during a collision. While it cannot be directly measured, it is derived by combining the observed number of events (N_{meas}) over a given time with the integrated luminosity ($\int L dt$), which encapsulates information about the particle beam's intensity and structure. The relationship is expressed as:

$$\sigma = \frac{N_{\text{meas}}}{\int L dt} \quad (3.1)$$

Although conceptually likened to a physical area where particles interact, the cross section is governed by quantum mechanical principles. It is commonly measured in barns ($1 \text{ b} = 10^{-24} \text{ cm}^2$), and in complex scenarios involving multiple possible interactions, the total cross section is the sum of all contributing partial cross sections ($\sigma = \sum \sigma_i$).

A simplified analogy views the cross section as a particle's "target area" for interaction. However, due to the stochastic nature of quantum processes, only specific combinations of energy and momentum result in interactions, making the cross section a probability-weighted quantity.

3.2 Selection Criteria

High-energy electron-positron collisions produce a variety of outcomes, classified as channels, based on the resultant particles. Analyzing these outcomes involves distinguishing between channels using measured data. However, due to overlapping variables, such differentiation can be challenging in extensive datasets. To address this, **selection cuts** are implemented—thresholds applied to key variables that enhance the distinction between channels.

For example, consider two classes of events, A and B, dependent on a parameter t . A threshold value t_{obs} is chosen to isolate most of the desired events in A to one side of the cut while minimizing contributions from B. This concept is illustrated in Figure 3.1. Selection cuts are refined iteratively, balancing accuracy and noise reduction.

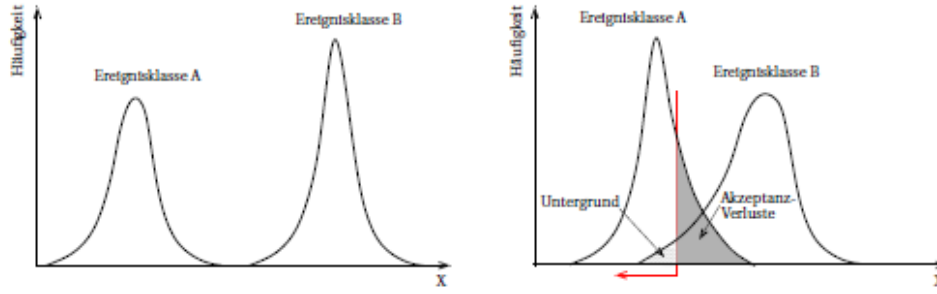


Figure 3.1: Illustration of selection cuts. (a) Distribution of two event classes A and B. (b) Application of a cut to minimize background and acceptance losses. Adapted from the original report.

Key metrics associated with selection cuts include:

- **Acceptance:** The fraction of desired events within a class that meet the cut criteria.
- **Efficiency:** The trade-off between acceptance and contamination from background events (undesired events relative to the total).

Monte Carlo (MC) simulations play a vital role in optimizing selection cuts. These simulations model collisions and account for acceptance losses and background interference, ensuring reliable estimates of critical quantities like the cross section. Without proper correction for these factors, the extracted parameters may deviate significantly from true values.

3.3 Fitting Techniques

Fitting is an analytical process used to interpret experimental data, particularly when desired physical parameters cannot be directly measured or theoretically computed. In the context of Z0-boson analysis, fitting helps model the relationship between measured quantities (e.g., observed events) and underlying physical parameters like the cross section.

Fitting involves selecting a mathematical model to describe the data and adjusting its free parameters to minimize discrepancies between observed and predicted values. The **least-squares method** is commonly employed, minimizing the sum of squared residuals, which represent the differences between measured and fitted data points. The choice of the model depends on the nature of the data; frequently used models include:

- **Exponential decay:** $f_1(x) = a \cdot e^{b \cdot x} + c$,
- **Gaussian distribution:** $f_2(x) = \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$,
- **Linear relationships:** $f_3(x) = a \cdot x + b$,

- **Breit-Wigner distribution** (critical for Z0-boson studies): $P(E) = \frac{\Gamma}{2\pi} \cdot \frac{1}{(E-E_0)^2 + (\Gamma/2)^2}$.

The Breit-Wigner function is particularly important for Z0 boson resonance studies, as it characterizes the distribution of cross sections around the resonance energy, allowing precise determination of parameters like decay width (Γ) and resonance mass (M).

Uncertainty quantification is integral to fitting. Measurement errors and the inherent variability of the fitting method itself contribute to the total uncertainty. The **chi-squared statistic** (χ^2) is widely used to evaluate the goodness of fit:

$$\chi^2 = \sum_{i=1}^N \left(\frac{y_i - f(x_i)}{\Delta y_i} \right)^2 \quad (3.2)$$

Here, y_i are the measured data points, $f(x_i)$ are the corresponding fitted values, and Δy_i are the uncertainties. To normalize for degrees of freedom in the fit, the reduced chi-squared (χ_{red}^2) is computed. Ideally, χ_{red}^2 should approach 1, with higher values suggesting a poorer fit.

Digital tools such as Python provide robust libraries for fitting routines, enabling efficient extraction of fitted parameters, uncertainties, and chi-squared values. These computational methods are invaluable for analyzing high-energy physics data, offering insights into Z0-boson properties and their alignment with theoretical predictions.

4 Preparation of Theoretical Aspects

4.1 Partial Decay Width

The decay of the Z0 boson into various fermion-antifermion pairs is governed by the electroweak interaction. The partial decay width (Γ_f) for a specific final state is calculated using the following formula:

$$\Gamma_f = N_c^f \frac{\sqrt{2}}{12\pi} G_F M_Z^3 \left[(g_V^f)^2 + (g_A^f)^2 \right] \quad (4.1)$$

where:

- $G_F = 1.166 \times 10^{-5} \text{ GeV}^{-2}$ is the Fermi coupling constant,
- N_c^f is the color factor (1 for leptons, 3 for quarks),
- $M_Z = 91.188 \text{ GeV}/c^2$ is the Z0 boson mass,
- g_V^f and g_A^f are the vector and axial-vector couplings for the specific fermion flavor.

The couplings are expressed in terms of the weak isospin (I_3^f), charge (Q^f), and the Weinberg angle ($\sin^2 \theta_W = 0.2312$) as:

$$g_V^f = I_3^f - 2Q^f \sin^2 \theta_W, \quad g_A^f = I_3^f$$

Using these formulas, the partial decay widths for different fermions are calculated as follows:

$$\Gamma_{e^-e^+} = \Gamma_{\mu^-\mu^+} = \Gamma_{\tau^-\tau^+} \approx 83.411 \text{ MeV}$$

$$\Gamma_{\nu_e\nu_e} = \Gamma_{\nu_\mu\nu_\mu} = \Gamma_{\nu_\tau\nu_\tau} \approx 165.884 \text{ MeV}$$

$$\Gamma_{u\bar{u}} = \Gamma_{c\bar{c}} \approx 285.414 \text{ MeV}, \quad \Gamma_{d\bar{d}} = \Gamma_{s\bar{s}} = \Gamma_{b\bar{b}} \approx 367.887 \text{ MeV}$$

The total decay width (Γ_Z) is the sum of the partial widths for all possible decay channels:

$$\Gamma_Z = \Gamma_{\text{had}} + \Gamma_{\text{charged leptons}} + \Gamma_{\text{invisible}}$$

where:

$$\Gamma_{\text{had}} = 1674.489 \text{ MeV}, \quad \Gamma_{\text{charged leptons}} = 250.232 \text{ MeV}, \quad \Gamma_{\text{invisible}} = 497.651 \text{ MeV}$$

$$\Gamma_Z \approx 2422.372 \text{ MeV}$$

4.2 Addition of Decay Channels

The addition of new fermion families introduces additional decay channels for the Z^0 boson, increasing its total decay width. For example:

- If a new lepton family is introduced, the total decay width increases to:

$$\Gamma_Z^{\text{new}} = \Gamma_Z + \Gamma_{e^-e^+} + \Gamma_{\nu_e\nu_e} \approx 2671.667 \text{ MeV}$$

- If a new quark family is introduced, the total decay width becomes:

$$\Gamma_Z^{\text{new}} = \Gamma_Z + \Gamma_{u\bar{u}} + \Gamma_{d\bar{d}} \approx 3075.673 \text{ MeV}$$

This corresponds to a relative increase of approximately 10% and 27%, respectively.

4.3 Expected Angular Distributions

The angular distribution of final-state particles in Z^0 -mediated processes depends on the interaction mechanism:

- For the s-channel, the differential cross section is symmetric and follows:

$$\frac{d\sigma}{d\Omega} \propto 1 + \cos^2 \theta$$

- For the t-channel, involving photon exchange, the distribution is dominated at small angles and follows:

$$\frac{d\sigma}{d\Omega} \propto \frac{1}{(1 - \cos \theta)^2}$$

These distributions are shown in Figure 4.1, highlighting the contrast between the two channels.

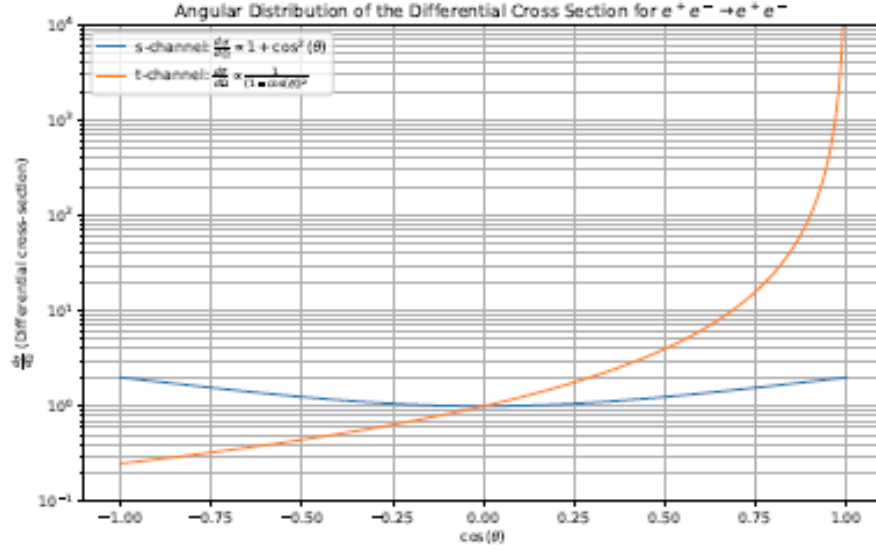


Figure 4.1: Angular distribution of the differential cross section for $e^+e^- \rightarrow e^+e^-$, showing the s-channel ($1 + \cos^2 \theta$) and t-channel ($1/(1 - \cos \theta)^2$) contributions.

4.4 Feynman Diagrams for s- and t-Channels

The interaction mechanisms in $e^+e^- \rightarrow e^+e^-$ are depicted in Figure 4.2. The s-channel represents Z^0 boson production, while the t-channel involves photon exchange. These diagrams are key to understanding the interplay of different contributions to the overall cross section.

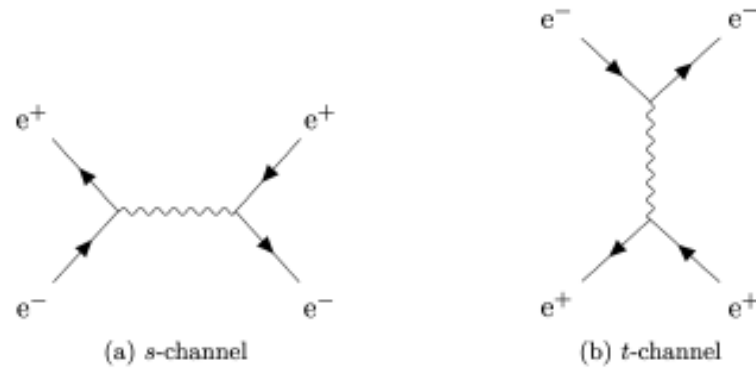


Figure 4.2: Feynman diagrams for $e^+e^- \rightarrow e^+e^-$: (a) s-channel Z^0 boson exchange, (b) t-channel photon exchange.

5 Analysis

5.1 Qualitative analysis of events

5.1.1 Features of events

The GROPE software (Graphic Reconstruction of OPAL Events) is an essential tool for visualizing and analyzing particle trajectories and interactions within the OPAL detector. It reconstructs particle paths and provides numerical values for key physical quantities, enabling a detailed study of events in the detector's sub-components. The quantities analyzed are as follows:

- $\text{Ctrk}(N) \equiv N_{\text{trk}}$: Number of particles detected in the central track chamber.
- $\text{Ctrk}(\text{Sump}) \equiv \text{Sump}$: Total momentum of charged particles in the central track chamber.
- $\text{Ecal}(N) \equiv N_{\text{ECAL}}$: Number of detections in the ECAL.
- $\text{Ecal}(\text{SumE}) \equiv E_{\text{ECAL}}$: Sum of energy deposited in the ECAL.
- $\text{Hcal}(N) \equiv N_{\text{HCAL}}$: Number of detections in the HCAL.
- $\text{Hcal}(\text{SumE}) \equiv E_{\text{HCAL}}$: Sum of energy deposited in the HCAL.
- $\text{Muon}(M) \equiv N_{\text{muon}}$: Number of detections in the muon chamber.

To gain familiarity with the detector's characteristics, four distinct pure data sets (screenshots), corresponding to the ee, mm, tt, and qq decay channels, are shown in below images. By studying these images, one can identify the distinct features that distinguish each channel. This understanding is essential for interpreting events in mixed data sets.

The identification of decay channels can be performed in two main steps: a visual method and a numerical method. Visually, GROPE uses unique markers to represent particle interactions. For example:

Yellow trapezoids signify electromagnetic showers. Red arrows represent muon tracks. Cyan solid lines display trajectories of charged particles within the central track chamber. Pink rectangles highlight detections in the HCAL. In contrast, the numerical method utilizes measurable

quantities such as the number of trajectories in the central track chamber or the number of detections in the muon chamber to distinguish between decay channels.

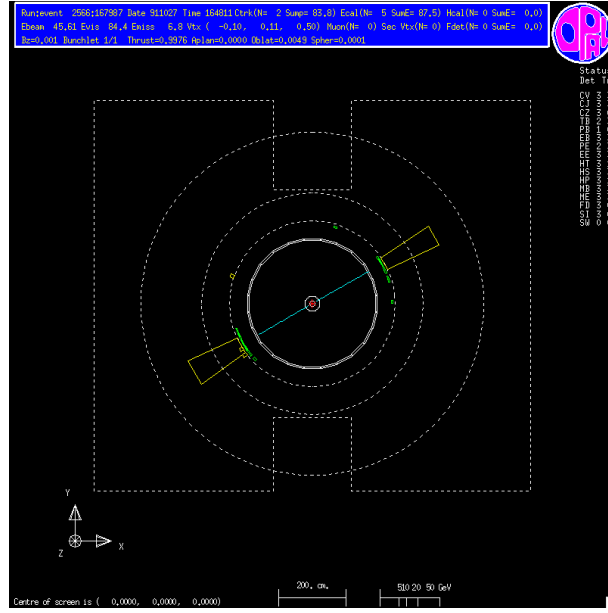


Figure 5.1: ee-channel: Two distinct trajectories are observed in the central track chamber (Ctrk), with the majority of energy being deposited in the electromagnetic calorimeter (ECAL).

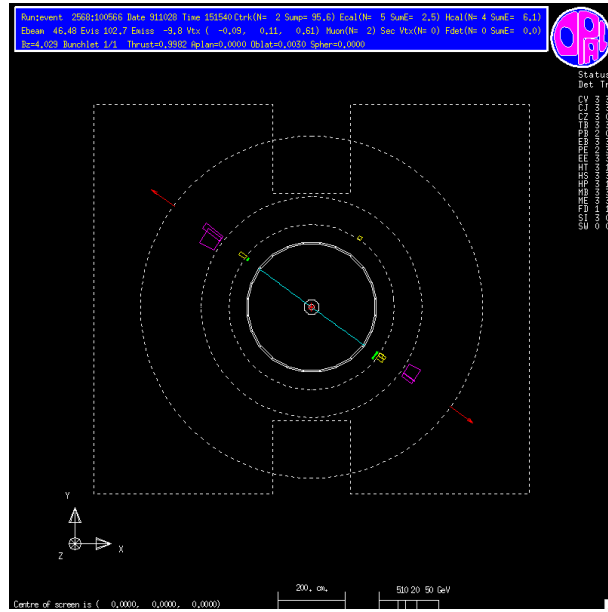


Figure 5.2: mm-channel: Two trajectories extend through both the ECAL and hadronic calorimeter (HCAL) before being detected in the muon chamber.

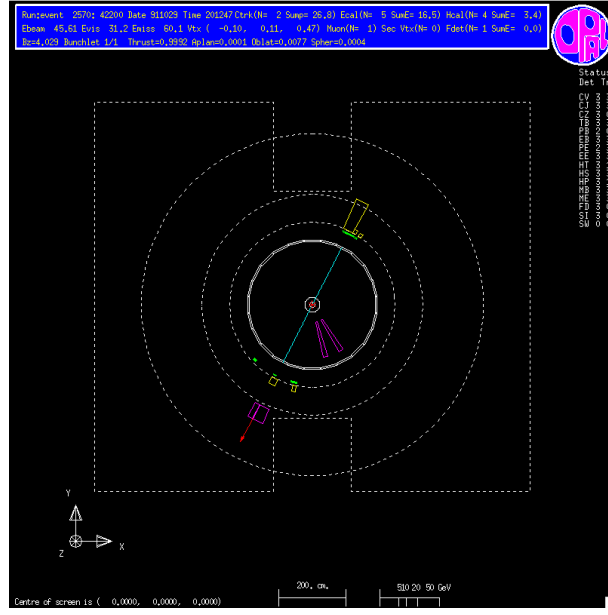


Figure 5.3: tt -channel: Multiple trajectories originate just outside the vertex chamber. These trajectories are less dispersed compared to the qq -channel. Energy deposition occurs primarily in the HCAL, with some energy also measured in the ECAL.

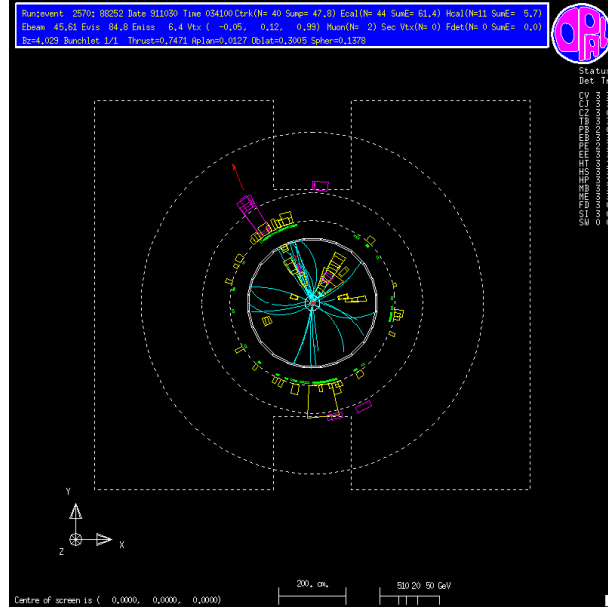


Figure 5.4: qq -channel: A large number of particle trajectories arise due to gluon-to-quark decays, leading to energy being recorded in both the ECAL and HCAL.

Below is a summary of some key particle characteristics observed in the OPAL detector for the various decay channels.

Scattering Channel: $e^+e^- \rightarrow e^+e^-$

Two straight tracks are observed in the Ctrk (central track chamber). The total momentum of charged particles (E_{sump}) is equivalent to the total visible energy (E_{visible}), indicating no energy loss due to neutrinos. Minimal energy deposition occurs in the HCAL (hadronic calorimeter). No detections are recorded in the MC (muon chamber).

Muon Channel: $e^+e^- \rightarrow \mu^+\mu^-$

Two straight tracks pass through the Ctrk, ECAL (electromagnetic calorimeter), HCAL, and MC. The total momentum is significantly high. Energy deposition is higher in the HCAL compared to the ECAL. Two detections are recorded in the Ctrk, ECAL, HCAL, and MC.

Tauon Channel: $e^+e^- \rightarrow \tau^+\tau^-$

Between two to five traces are observed in the Ctrk. The total momentum is lower due to energy being transferred to invisible neutrinos. Energy deposition occurs in both the ECAL and HCAL. Detections are recorded in the Ctrk, ECAL, HCAL, and MC.

Quark Channel: $e^+e^- \rightarrow q\bar{q}$

Typically, 10 or more traces are observed in the Ctrk. The total momentum loss is higher compared to the e^+e^- and $\mu^+\mu^-$ channels. Jets are produced inside the vertex chamber. Energy is deposited in both the ECAL and HCAL, with the HCAL showing higher energy deposition.

5.1.2 Evaluation of 10 random GROPE images

To gain familiarity with identifying particle decay channels using the GROPE software, 10 randomly selected event images were analyzed. Each image was visually inspected to evaluate the characteristic features of the particle trajectories, energy deposition patterns, and detector responses. Based on these observations, the events were classified into one of the four decay channels: $e^+e^- \rightarrow e^+e^-$ (scattering), $e^+e^- \rightarrow \mu^+\mu^-$ (muon channel), $e^+e^- \rightarrow \tau^+\tau^-$ (tauon channel), or $e^+e^- \rightarrow q\bar{q}$ (quark channel).

The classification criteria were derived from the standard features observed in each channel, such as the number of tracks in the Ctrk, energy deposition in the ECAL and HCAL, and detections in the MC.

Table 5.1: Event Classification of 10 random GROPE images

Image	N-Ctr	Sump	Evis	Emis	N-Ecal	E-Ecal	N-Hcal	E-Hcal	N- μ	Channel
31	2	90.5	90.8	0.4	8	93.2	0	0.0	0	ee
39	2	69.4	70.2	21.0	5	92.9	1	0.2	0	ee
24	2	95.2	103.1	-11.9	4	1.3	6	7.9	2	$\mu\mu$
23	33	24.6	48.0	43.2	34	43.4	8	14.8	0	qq
14	2	89.1	89.1	2.1	6	89.7	0	0.0	0	ee
0	19	39.5	83.9	7.2	31	44.3	13	15.6	2	qq
20	45	65.3	80.3	10.8	52	51.2	7	2.4	1	qq
3	2	90.8	95.0	-3.8	5	1.4	5	4.1	2	$\mu\mu$
30	2	106.2	110.1	-18.9	10	2.6	4	3.6	2	$\mu\mu$
2	2	95.7	96.4	-5.2	9	93.4	0	0.0	0	ee

Table 5.2: 4x4 table describing variables that are important for separating each pair of channels

Decay Mode	$Z^0 \rightarrow e^+e^-$	$Z^0 \rightarrow \mu^+\mu^-$	$Z^0 \rightarrow \tau^+\tau^-$	$Z^0 \rightarrow q\bar{q}$
$Z^0 \rightarrow e^+e^-$	—	E_{ecal}	$E_{\text{ecal}}, E_{\text{hcal}}$	$N_{\text{charged}}, E_{\text{hcal}}$
$Z^0 \rightarrow \mu^+\mu^-$	E_{ecal}	—	P_{charged}	$N_{\text{charged}}, E_{\text{hcal}}$
$Z^0 \rightarrow \tau^+\tau^-$	$E_{\text{ecal}}, E_{\text{hcal}}$	P_{charged}	—	$N_{\text{charged}}, E_{\text{ecal}}$
$Z^0 \rightarrow q\bar{q}$	$N_{\text{charged}}, E_{\text{hcal}}$	$N_{\text{charged}}, E_{\text{ecal}}$	$N_{\text{charged}}, E_{\text{ecal}}$	—

5.2 Quantitative analysis of events

5.2.1 Analysis of Histograms for Different Z^0 -Boson Decay Channels

In this section, we analyze the event distributions for different Z^0 -Boson decay channels: e^+e^- (electron channel), $\mu^+\mu^-$ (muon channel), $\tau^+\tau^-$ (tau channel), and $q\bar{q}$ (quark channel). Each histogram illustrates key properties and features of the decay processes in simulated Monte Carlo (MC) datasets.

Electron Channel (e^+e^-)

The $Z^0 \rightarrow e^+e^-$ channel exhibits a discrete distribution of charged tracks with distinct peaks at 2 and 4, corresponding to electron-positron pair production, as shown in Figure 5.5. Higher track multiplicities are suppressed. The momentum sum of charged tracks peaks at low values, indicating the collinear nature of the electron-positron pairs, while the logarithmic momentum distribution is concentrated at higher values, reflecting the back-to-back emission near the Z^0 -resonance. A sharp peak around 90 GeV in the electromagnetic calorimeter (ECAL) confirms significant energy deposition by electrons and positrons. In contrast, energy deposition in the hadronic calorimeter (HCAL) is negligible, consistent with minimal hadronic interaction. The angular distribution reveals a highly collinear back-to-back emission pattern along the beamline, characteristic of electron-positron decays.

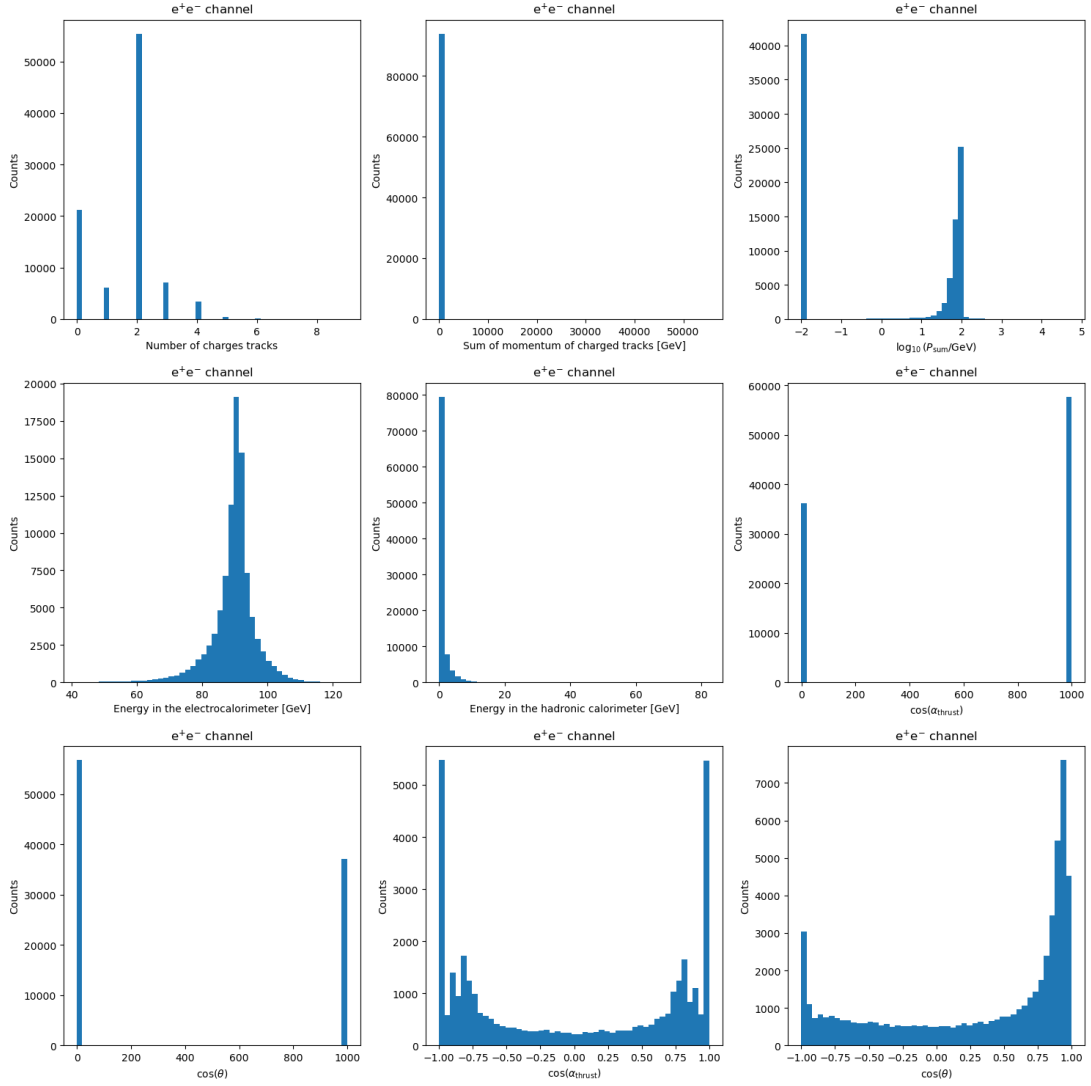


Figure 5.5: e^+e^- channel: Histograms illustrating key event properties.

Muon Channel ($\mu^+\mu^-$)

In the $Z^0 \rightarrow \mu^+\mu^-$ channel, the number of charged tracks peaks at 2, reflecting clean muon-antimuon final states (Figure 5.6). The momentum distribution resembles that of the electron channel, with a peak at low values and a concentrated logarithmic momentum distribution. Minimal energy deposition is observed in the ECAL since muons interact weakly with the electromagnetic calorimeter. However, the HCAL registers small but non-zero energy deposition, consistent with the penetrating nature of muons through dense detector layers. The angular distribution remains symmetric and is centered around 0, indicating a back-to-back emission pattern in the transverse plane, similar to the electron channel.

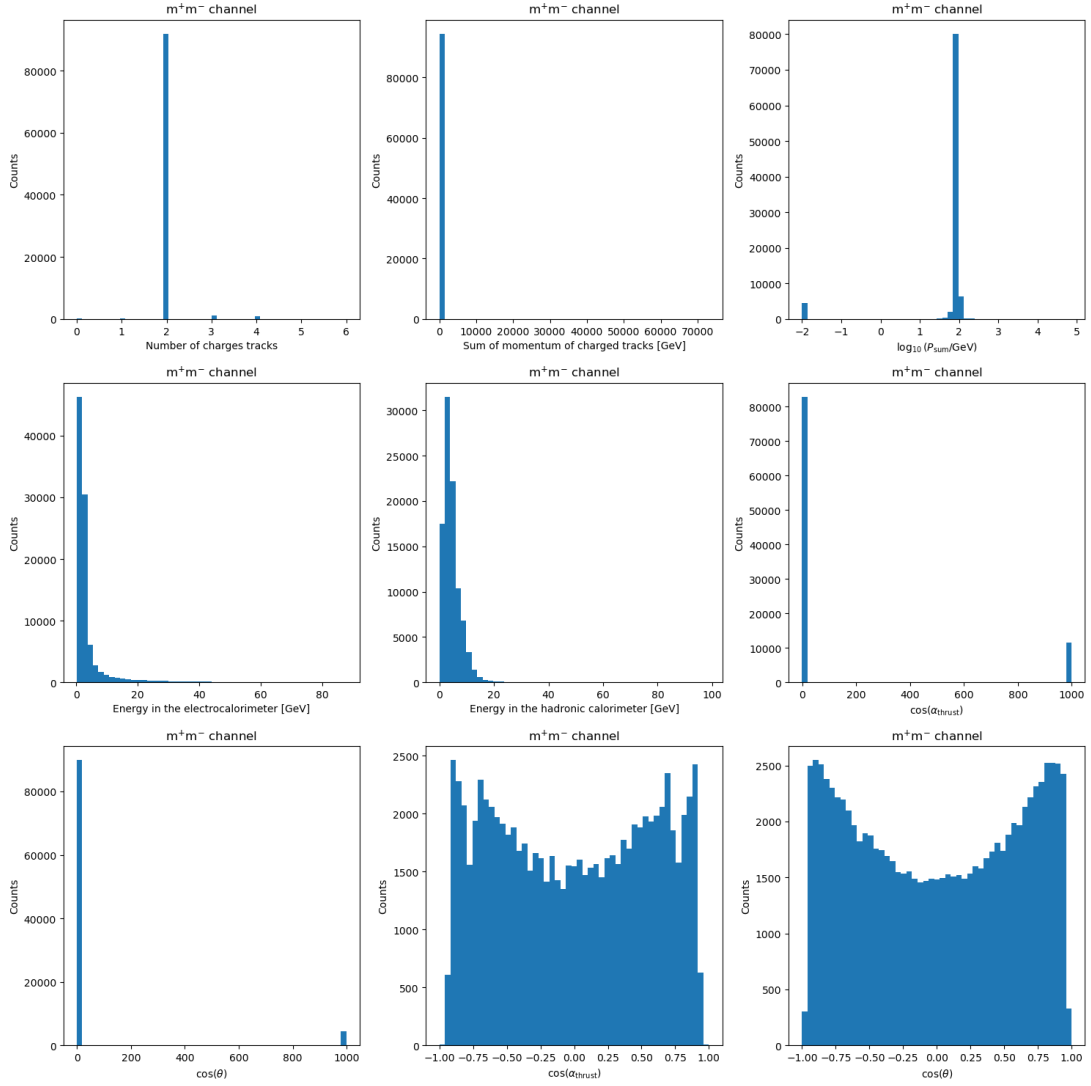


Figure 5.6: $\mu^+\mu^-$ channel: Histograms illustrating key event properties.

Tau Channel ($\tau^+\tau^-$)

The $Z^0 \rightarrow \tau^+\tau^-$ channel exhibits a broader charged track distribution compared to the electron and muon channels due to the secondary decays of tau leptons (Figure 5.7). Energy deposition occurs in both the ECAL and HCAL, with moderate peaks in the ECAL and significant contributions in the HCAL, particularly from hadronic tau decays. The angular distribution is less collinear than in the electron and muon channels, indicating a wider spread caused by the more complex decay processes of tau leptons. These observations highlight the intermediate nature of tau decays, which are neither as clean as leptonic decays nor as broad as hadronic events.

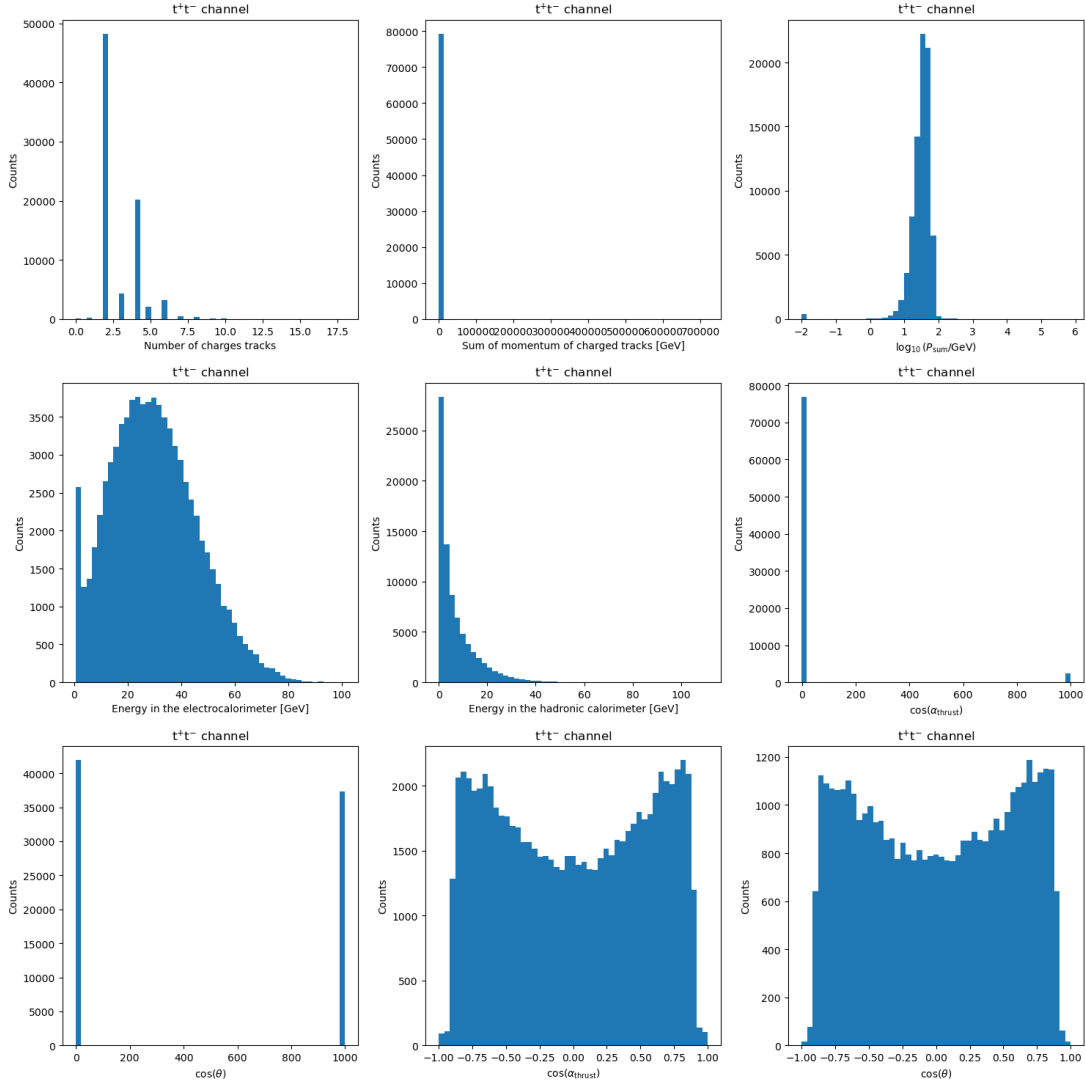


Figure 5.7: $\tau^+\tau^-$ channel: Histograms illustrating key event properties.

Quark Channel ($q\bar{q}$)

In the $Z^0 \rightarrow q\bar{q}$ channel, the number of charged tracks peaks around 20, reflecting the high multiplicity of hadronic jets resulting from quark hadronization (Figure 5.8). The momentum sum of charged tracks is broadly distributed, and the logarithmic momentum spans a wide range, consistent with the production of multi-particle jets. Significant energy deposition occurs in both the ECAL and HCAL, with the HCAL registering substantial energy due to hadronic interactions. The angular distribution is broad and isotropic, reflecting the nature of multi-particle production in quark fragmentation processes. These characteristics distinguish the quark channel from the leptonic channels, emphasizing the hadronic nature of the decay.

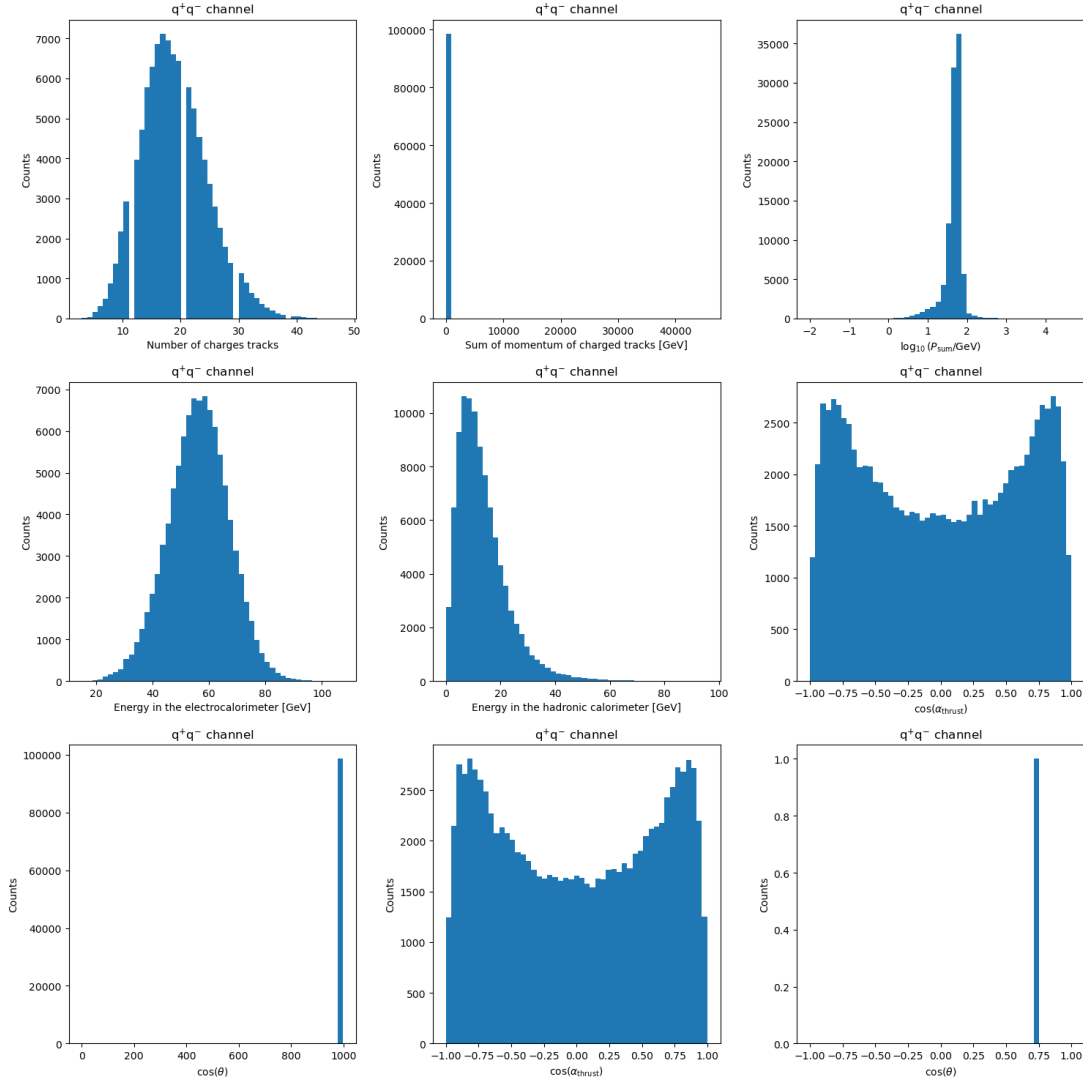


Figure 5.8: $q\bar{q}$ channel: Histograms illustrating key event properties.

5.2.2 Comparative Analysis of Z^0 -Boson Decay Channels and Plots

The e^+e^- and $\mu^+\mu^-$ channels consistently show sharp peaks at 2 charged tracks, low total momentum, and minimal energy deposition in the HCAL, with e^+e^- exhibiting significant energy in the ECAL near 90 GeV. The $\tau^+\tau^-$ channel lies in between, displaying a broader distribution of charged tracks, moderate momentum, and energy spread across both the ECAL and HCAL due to tau decays. In contrast, the $q\bar{q}$ channel shows the most distinct behavior, with a large number of charged tracks, high total momentum, and significant energy deposition primarily in the HCAL, reflecting hadronic jet activity. Angular distributions further emphasize the collimated nature of leptonic decays (e^+e^- and $\mu^+\mu^-$) compared to the isotropic patterns seen in $q\bar{q}$, with $\tau^+\tau^-$ showing an intermediate spread.

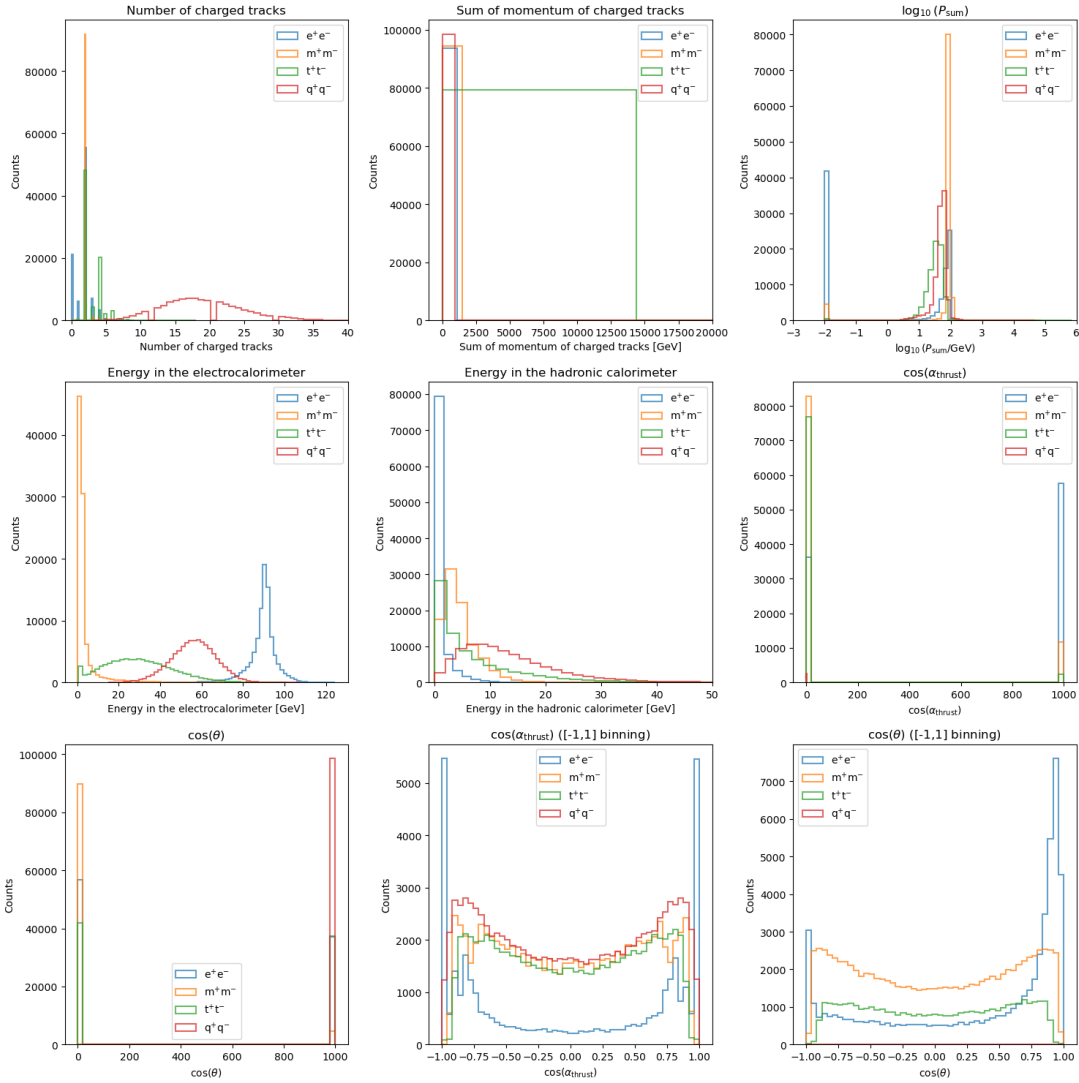


Figure 5.9: Comparative analysis of key variables for Z^0 -Boson decay channels: e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$, and $q\bar{q}$.

For P_{sum} , the logarithm $\log_{10}$ was used to compress the data range for better visualization and analysis, but 0.01 was added to avoid the undefined $\log_{10}(0)$ when P_{sum} is zero or very small. Regarding the angles, the constraint $-1 < \cos(\theta) < 1$ ensures physical validity, as $\cos(\theta)$ values outside this range are mathematically impossible in real angular measurements. Events with unphysical angles likely result from numerical errors, mis-reconstructed events, or artifacts in the data and are typically excluded from analysis to maintain accuracy.

5.2.3 Separating s-t channels

In this exercise, we analyzed the angular distribution of events in the e^+e^- channel to distinguish between the s-channel and t-channel contributions. The angular distribution was described using

the function:

$$a(1 + \cos^2 \theta) + b(1 - \cos \theta)^2 \quad (5.1)$$

where a and b are free parameters fitted to the data. A safe cut on $\cos(\theta)$ was applied, selecting the region $-0.9 < \cos(\theta) < 0.5$ to ensure physical validity.

The fit yielded the parameters:

$$a = 7881.059 \pm 88.17 \quad \text{and} \quad b = 827.737 \pm 78.86.$$

Using these parameters, we calculated the fraction of s-channel events within the constrained region $\cos(\theta) \in [-0.9, 0.5]$, obtaining a value of approximately 0.915. The total number of s-channel events was also integrated over $\cos(\theta) \in [-1, 1]$, resulting in:

$$13276.957 \pm 1.47.$$

The final plot (Figure 5.10) clearly shows the total event distribution (dashed black line), the s-channel contribution (cyan curve), and the t-channel contribution (magenta curve), along with the fitted function. This analysis effectively separates the s-channel and t-channel processes based on their distinct angular dependencies.

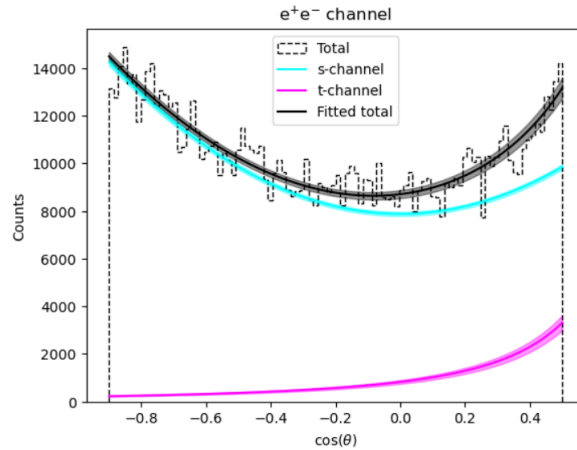


Figure 5.10: Angular distribution in the e^+e^- channel. The total event distribution (dashed black line) is distributed into the s-channel (cyan curve) and t-channel (magenta curve) contributions. The fitted total function is shown in black.

5.2.4 Acceptance Matrix and Corrected Acceptance Matrix

Now we derive and analyze the survival fraction matrix, which quantifies how well the defined selection cuts separate the different Z^0 -Boson decay channels: $Z^0 \rightarrow e^+e^-$ (electron channel), $Z^0 \rightarrow \mu^+\mu^-$ (muon channel), $Z^0 \rightarrow \tau^+\tau^-$ (tau channel), and $Z^0 \rightarrow q\bar{q}$ (quark channel). Monte Carlo (MC) simulations were performed with 10^5 total events generated per channel. The survival fraction matrix, correction for the electron channel, and ideal behavior are discussed below.

Selection Cuts

Using the predefined cuts for each decay channel, masks are applied to filter events based on their kinematic properties. The cuts are:

- e^+e^- **Channel:** $(\cos\theta > -0.9) \wedge (\cos\theta < 0.5) \wedge (N_{\text{charged}} < 6) \wedge (E_{\text{HCAL}} < 17) \wedge (E_{\text{ECAL}} > 55)$,
- $\mu^+\mu^-$ **Channel:** $(1.8 < N_{\text{charged}} < 5) \wedge (E_{\text{HCAL}} < 10) \wedge (E_{\text{ECAL}} < 10)$,
- $\tau^+\tau^-$ **Channel:** $(1.8 < N_{\text{charged}} < 8) \wedge (5 < E_{\text{ECAL}} < 80)$,
- $q\bar{q}$ **Channel:** $(N_{\text{charged}} > 7) \wedge (40 < E_{\text{ECAL}} < 80)$.

Survival Fraction Matrix

The survival fraction matrix A_{raw} represents the fraction of events in channel j that survive the selection cuts defined for channel i . Mathematically, it is expressed as:

$$A_{ij} = \frac{N_{\text{survived},i}(j)}{N_{\text{total}}(j)},$$

where $N_{\text{survived},i}(j)$ is the number of events in channel j that pass the cuts for channel i , and $N_{\text{total}}(j)$ is the total number of simulated events for channel j .

The raw survival fraction matrix obtained is:

$$A_{\text{raw}} = \begin{bmatrix} 0.210688 & 0.000053 & 0.035032 & 0.000000 \\ 0.000000 & 0.744832 & 0.039968 & 0.000000 \\ 0.005330 & 0.061495 & 0.737067 & 0.001441 \\ 0.000000 & 0.000000 & 0.000429 & 0.625884 \end{bmatrix}.$$

Here, the diagonal elements A_{ii} represent the fraction of events correctly identified within their respective channels, for example:

Electron channel: 21.07%, Muon channel: 74.48%, Tau channel: 73.71%, Quark channel: 62.59%.

The off-diagonal elements $A_{ij}(i \neq j)$ indicate cross-channel contamination, such as 3.5% of $\tau^+\tau^-$ events surviving the e^+e^- -cuts and 4% surviving the $\mu^+\mu^-$ -cuts. In an ideal scenario, the survival fraction matrix would be diagonal with $A_{ii} = 1$ for each channel and $A_{ij} = 0$ for $i \neq j$. However, detector limitations and misidentifications introduce small off-diagonal elements.

Correction for the e^+e^- -Channel

To account for missing s-channel events in the e^+e^- -channel caused by angular selection cuts, we apply a correction factor s , defined as:

$$s = \frac{\text{fracSChannel} \cdot \text{Total MC Events}}{\text{Total S-Channel Events}} \quad (5.2)$$

With the given values:

$\text{fracSChannel} = 0.9159$, $\text{Total S-Channel Events} = 20984.6498$, $\text{Total MC Events} = 100000$,

the correction factor is calculated as:

$$s = 0.9159 \cdot \frac{100000}{20984.6498} = 4.3646.$$

The e^+e^- -row of the survival matrix is scaled by $s = 4.3646$ to obtain the corrected survival fraction matrix.

Corrected Survival Fraction Matrix

After applying the correction factor, the updated matrix becomes:

$$A_{\text{corrected}} = \begin{bmatrix} 0.919575 & 0.000231 & 0.152900 & 0.000000 \\ 0.000000 & 0.744832 & 0.039968 & 0.000000 \\ 0.005330 & 0.061495 & 0.737067 & 0.001441 \\ 0.000000 & 0.000000 & 0.000429 & 0.625884 \end{bmatrix}.$$

The diagonal survival fraction for the e^+e^- -channel increases to 91.96%, accurately reflecting the corrected s-channel contribution. Off-diagonal contamination remains negligible, indicating that the selection cuts effectively suppress background events.

Conclusion

The corrected survival fraction matrix demonstrates that the stringent cuts applied to the e^+e^- -channel are effective after accounting for missing s-channel contributions. Ideally, the survival matrix would be perfectly diagonal, but practical limitations lead to small off-diagonal elements. Evaluating the matrix quality requires minimizing contamination and ensuring the efficiency of event selection. This analysis provides a robust foundation for accurate separation of decay channels and further statistical studies.

5.3 Getting the physical results

5.3.1 Estimating the Number of Real Data Events after Cuts

After learning from the Monte Carlo simulations and establishing reliable selection cuts for the four decay channels ($Z^0 \rightarrow e^+e^-$, $Z^0 \rightarrow \mu^+\mu^-$, $Z^0 \rightarrow \tau^+\tau^-$, and $Z^0 \rightarrow q\bar{q}$), we now apply these cuts to real experimental data. Unlike simulations, the real data does not inherently carry information about the originating decay channel for each event. Therefore, the selection cuts allow us to estimate the number of detected events that belong to each channel.

Real Data Event Distribution After Cuts

The analysis proceeds as follows:

Energy Bins: The real data is divided into multiple energy bins k , defined by the range:

$$E_k = [44, 44.5, 45, 45.4, 45.8, 46.2, 46.7, 47.2] \text{ GeV}.$$

These bins correspond to the energy ranges where events are detected and allow for a structured analysis across varying energies close to the Z^0 -resonance.

Event Counting: The number of events surviving each set of cuts is counted for all energy bins. This is done by applying the masks for each channel and energy bin to the real data. The filtered events $N_{\text{det},j,k}$ represent the number of detected events in energy bin k for channel j .

Detected Events Matrix

The resulting matrix N_{det} gives the number of real data events surviving each set of cuts across the energy bins:

$$N_{\text{det}} = \begin{bmatrix} 102 & 93 & 242 & 2176 \\ 193 & 194 & 348 & 4144 \\ 314 & 392 & 609 & 8225 \\ 3356 & 4947 & 6815 & 104028 \\ 335 & 552 & 762 & 11135 \\ 206 & 373 & 567 & 7776 \\ 166 & 310 & 442 & 6288 \end{bmatrix}.$$

Here:

- Each row corresponds to an energy bin k (from 44 GeV to 47.2 GeV).
- Each column corresponds to a specific decay channel (e, μ, τ, q).
- $N_{\text{det},j,k}$ is the number of events that passed the cuts for channel j in energy bin k .

Observations: The $q\bar{q}$ -channel consistently has the largest number of detected events across all energy bins. This result aligns with theoretical expectations, as the Z^0 -Boson predominantly decays into hadronic jets ($q\bar{q}$) due to the strong interaction. The $\tau^+\tau^-$ -channel ($t\bar{t}$) shows a moderate number of surviving events, reflecting its more complex decay products that still fit within the defined selection cuts. The e^+e^- (ee) and $\mu^+\mu^-$ (mm) channels have relatively fewer events, consistent with their lower branching ratios compared to the hadronic decay channel.

5.3.2 Estimation of the Initial Number of Events in Each Channel

The detected event matrix N_{det} obtained in the previous step does not directly represent the true initial number of events in each decay channel. The selection cuts applied earlier result in some events being excluded and introduce cross-contamination between channels. To account for this, we use the acceptance matrix A , which quantifies how events from one channel can survive cuts defined for another.

Estimating N_{init}

To estimate the initial number of events N_{init} in each channel, we solve the following matrix equation:

$$N_{\text{det},k} = A \cdot N_{\text{init},k}.$$

Here:

- $N_{\text{det},k}$ is the vector of detected events across all channels for energy bin k ,
- A is the corrected acceptance matrix obtained earlier,
- $N_{\text{init},k}$ is the vector of initial events to be estimated.

Rearranging for $N_{\text{init},k}$, we obtain:

$$N_{\text{init},k} = A^{-1} \cdot N_{\text{det},k} \quad (5.3)$$

where A^{-1} is the inverse of the acceptance matrix.

Inverted Acceptance Matrix

Using the corrected acceptance matrix A , its inverse A^{-1} is calculated as:

$$A^{-1} = \begin{bmatrix} 1.0888 & 0.0184 & -0.2269 & 0.0005 \\ 0.0004 & 1.3486 & -0.0732 & 0.0002 \\ -0.0079 & -0.1127 & 1.3645 & -0.0031 \\ 0.0000 & 0.0001 & -0.0009 & 1.5977 \end{bmatrix}.$$

This matrix is crucial for correcting the detected events, as it reverses the effects of both the selection cuts and cross-channel contamination.

Estimation of N_{init} : Initial Number of Events

By applying the inverse acceptance matrix A^{-1} to the N_{det} matrix, we estimate N_{init} , the initial number of events in each channel across the energy bins:

$$N_{\text{init}} = \begin{bmatrix} 59.00 & 108.11 & 312.08 & 3476.47 \\ 136.92 & 236.93 & 438.44 & 6620.73 \\ 215.22 & 485.59 & 758.49 & 13140.89 \\ 2253.21 & 6191.65 & 8388.29 & 166203.96 \\ 207.84 & 690.67 & 939.92 & 17790.19 \\ 106.58 & 462.92 & 705.58 & 12423.54 \\ 89.45 & 386.84 & 547.11 & 10046.22 \end{bmatrix}.$$

5.3.3 Estimation of the Cross Sections

Having determined the initial number of events N_{init} for each decay channel across the energy bins, we now estimate the corresponding cross sections. The cross section quantifies the likelihood of the Z^0 -Boson decaying into each specific channel as a function of energy.

The cross section for channel i at energy bin k is given by:

$$\sigma_i(E_k) = \frac{N_{i,k}}{L_k} + \eta_{i,k} \quad (5.4)$$

where:

- $N_{i,k}$: The estimated initial number of events for channel i at energy E_k ,
- L_k : The integrated luminosity at energy E_k ,
- $\eta_{i,k}$: The radiation corrections for leptonic (η_{lep}) and hadronic (η_{had}) channels.

Cross Section Results

The computed cross sections for the four decay channels are as follows:

e^+e^- -Channel: The cross section for the e^+e^- -channel is:

$$\sigma_{ee} = [0.621, 1.127, 2.917, 20.163, 1.984, 1.034, 0.868] [\text{GeV}^{-2}].$$

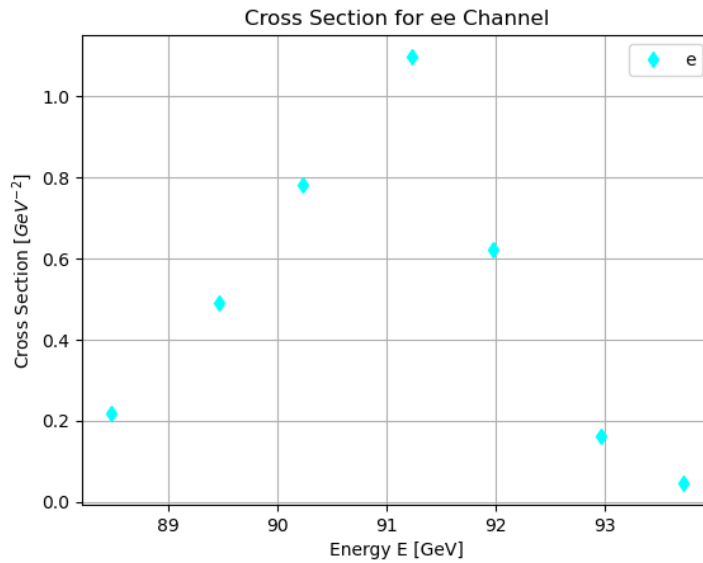


Figure 5.11: Cross section for the e^+e^- -channel as a function of energy.

$\mu^+\mu^-$ -**Channel:** The cross section for the $\mu^+\mu^-$ -channel is:

$$\sigma_{mm} = [1.103, 2.482, 4.543, 36.018, 6.582, 4.423, 3.697] [\text{GeV}^{-2}].$$

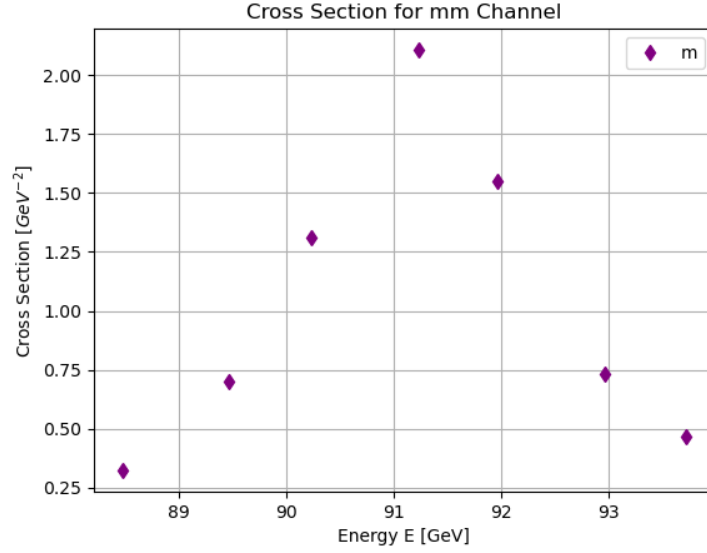


Figure 5.12: Cross section for the $\mu^+\mu^-$ -channel as a function of energy.

$\tau^+\tau^-$ -**Channel:** The cross section for the $\tau^+\tau^-$ -channel is:

$$\sigma_{tt} = [3.101, 4.132, 6.710, 48.356, 7.998, 6.059, 5.144] [\text{GeV}^{-2}].$$

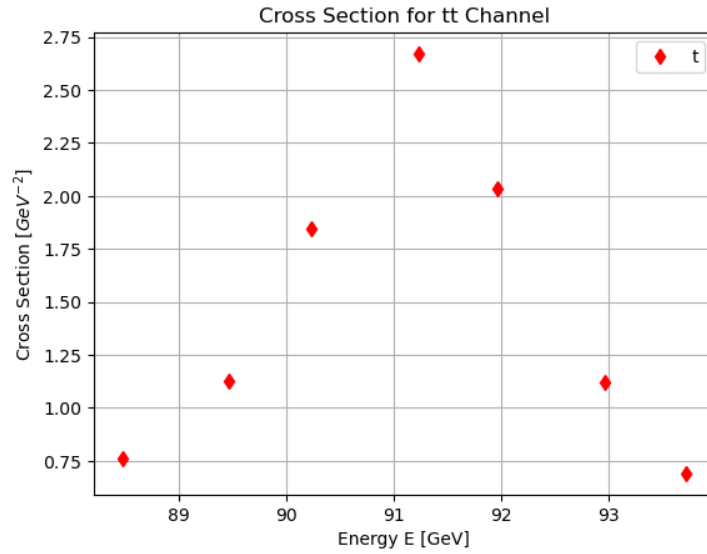


Figure 5.13: Cross section for the $\tau^+\tau^-$ -channel as a function of energy.

$q\bar{q}$ -Channel: The cross section for the $q\bar{q}$ -channel is:

$$\sigma_{q\bar{q}} = [30.218, 57.845, 88.534, 236.402, 65.913, 48.750, 39.875] [\text{GeV}^{-2}].$$

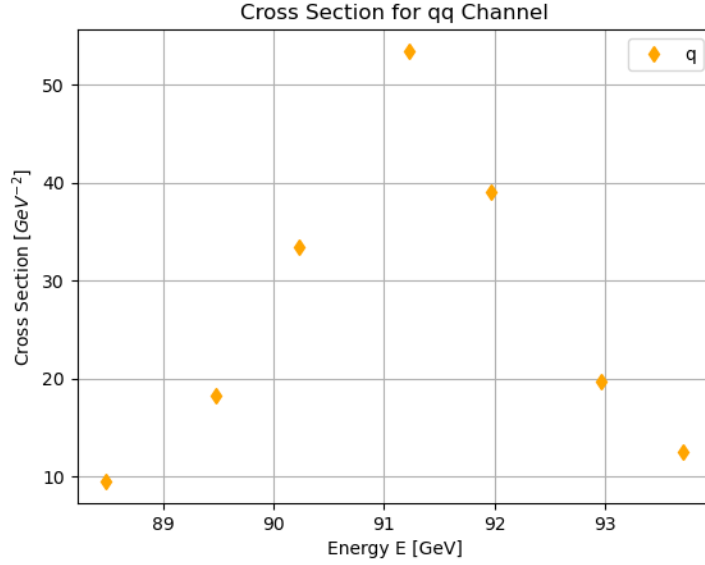


Figure 5.14: Cross section for the $q\bar{q}$ -channel as a function of energy.

Visual Representation of the Cross Sections

The cross sections for all channels are displayed as a function of energy. The combined plot provides a comprehensive comparison of the cross section behavior across the four decay channels:

- **e^+e^- -Channel:** The cross section peaks at intermediate energies near the Z^0 -resonance and decreases at both lower and higher energies, reflecting the strong preference for e^+e^- -pair production at the Z^0 -pole.
- **$\mu^+\mu^-$ -Channel:** The behavior closely follows that of the e^+e^- -channel due to the leptonic nature of the muonic decay.
- **$\tau^+\tau^-$ -Channel:** This channel exhibits a broader peak compared to the e^+e^- and $\mu^+\mu^-$ -channels, attributed to the additional phase space available for tau decays.
- **$q\bar{q}$ -Channel:** The hadronic channel shows the largest cross section across all energy bins, with a sharp peak near the Z^0 -resonance, reflecting the dominant branching ratio of the Z^0 -Boson decaying into quarks.

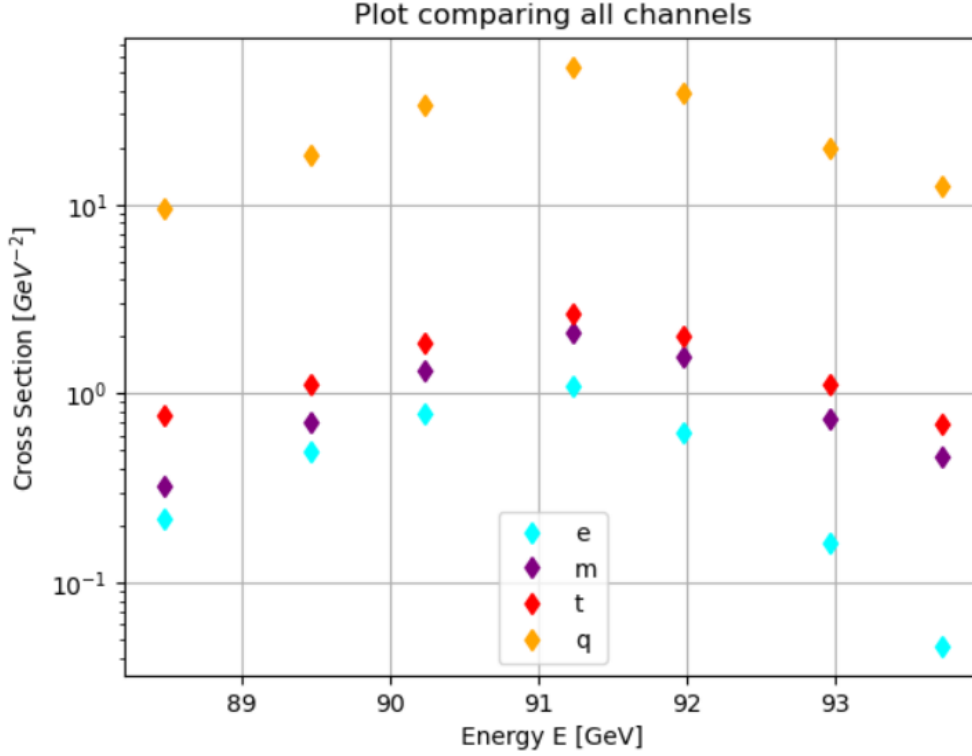


Figure 5.15: Combined cross section plot comparing all channels. The hadronic channel ($q\bar{q}$) shows the highest cross section, while leptonic channels peak similarly near the Z^0 -resonance.

5.3.4 Fitting the Breit-Wigner Function to Cross Sections

To determine the physical parameters of the Z^0 -Boson, we fit the measured cross-section data for each decay channel (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$, $q\bar{q}$) using the Breit-Wigner function. This function describes the behavior of the cross section near a particle's resonance energy, such as that of the Z^0 -Boson.

The Breit-Wigner Function

The Breit-Wigner formula is expressed as:

$$\sigma(E) = \frac{a}{\frac{4(E-b)^2}{c^2} + \left(\frac{E}{b}\right)^2}, \quad (5.5)$$

where:

- a : Maximum cross section at the resonance peak,

- b : Resonance energy, which corresponds to the mass of the Z^0 -Boson,
- c : Decay width (Γ_Z), representing the spread of the resonance peak.

Fitting Results

Using the provided code, the Breit-Wigner function was fitted to the measured cross-section data for each channel. The optimized parameters a , b , and c for the four channels are:

Table 5.3: Fitted Breit-Wigner parameters for each channel.

Channel	a (GeV^2)	b (GeV)	c (GeV)
e^+e^-	1.189	90.928	2.077
$\mu^+\mu^-$	2.108	91.247	2.519
$\tau^+\tau^-$	2.627	91.207	3.052
$q\bar{q}$	52.939	91.247	2.612

Visual Representation of Breit-Wigner Fits

The Breit-Wigner fits for all channels are shown below:

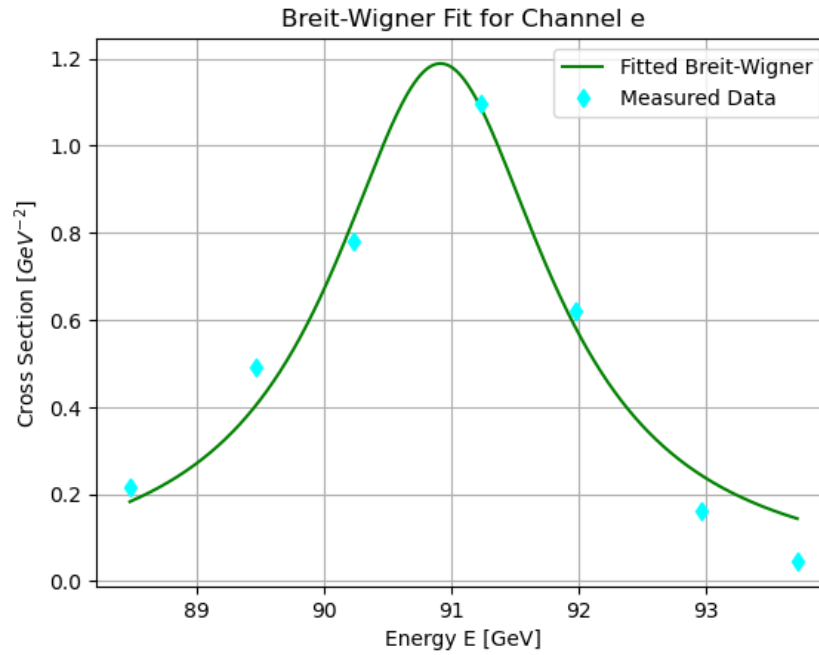


Figure 5.16: Breit-Wigner fit for the e^+e^- -channel. The measured data is overlaid with the fitted function.

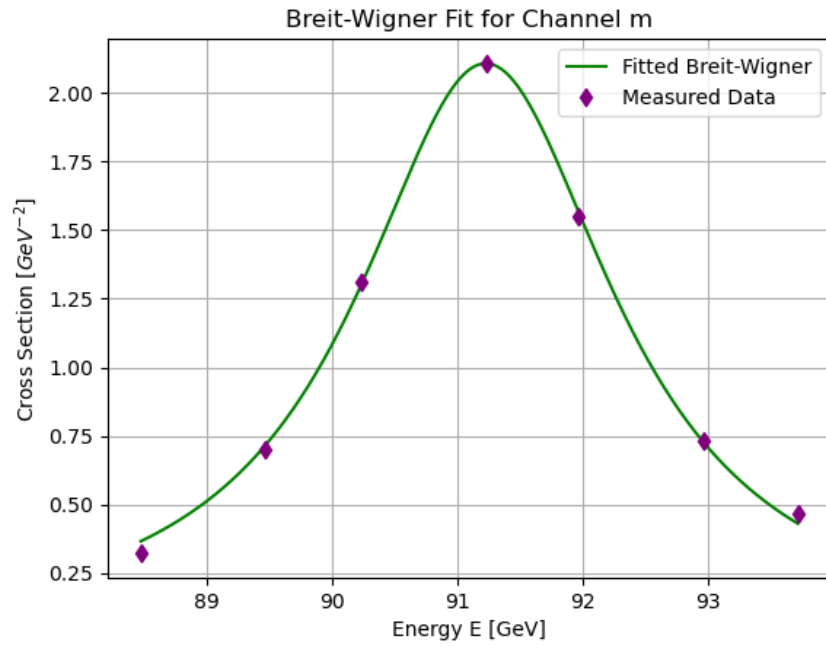


Figure 5.17: Breit-Wigner fit for the $\mu^+\mu^-$ -channel. The measured data is overlaid with the fitted function.

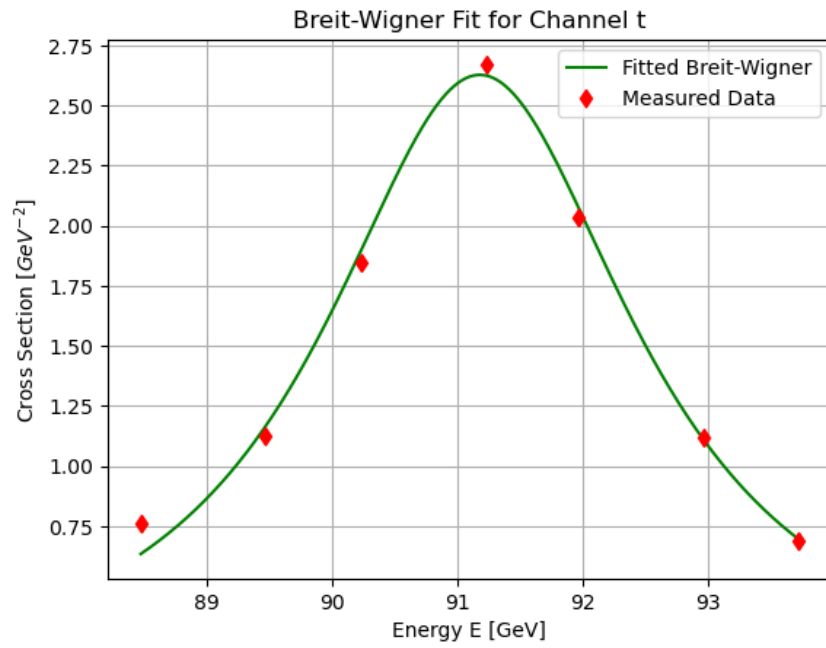


Figure 5.18: Breit-Wigner fit for the $\tau^+\tau^-$ -channel. The measured data is overlaid with the fitted function.

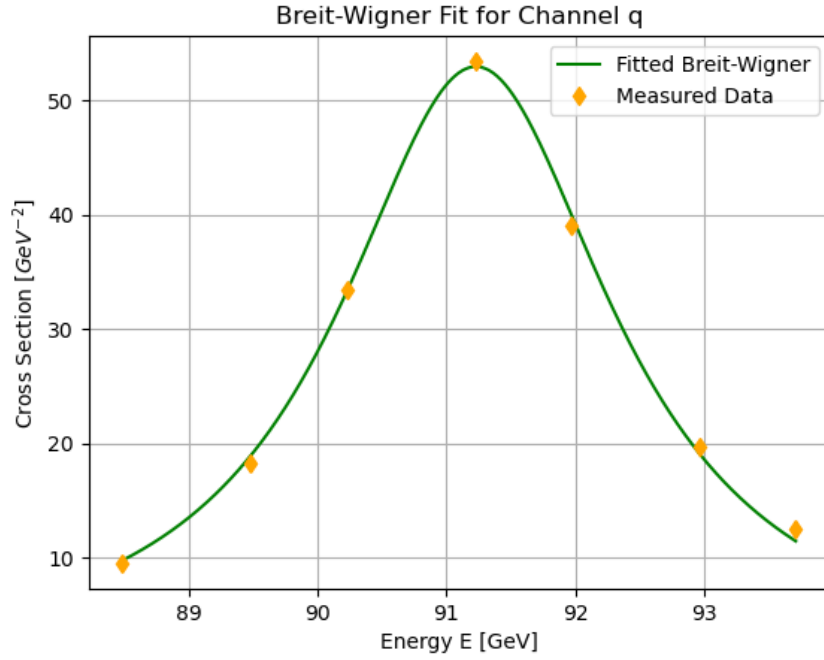


Figure 5.19: Breit-Wigner fit for the $q\bar{q}$ -channel. The measured data is overlaid with the fitted function.

Interpretation of the Parameters

Cross Section (a): The $q\bar{q}$ -channel ($a = 52.939$) has the largest maximum cross section, reflecting the dominant decay probability of the Z^0 -Boson into hadronic states. The leptonic channels (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$) have significantly smaller cross-section values, consistent with their lower branching ratios.

Resonance Energy (b): The fitted b -values for all channels are very close to 91.2 GeV, aligning with the known mass of the Z^0 -Boson ($m_Z \sim 91.19$ GeV). Slight deviations in the e^+e^- -channel ($b = 90.927$) may arise due to statistical uncertainties.

Decay Width (c): The decay width c represents the spread of the resonance peak, related to the Z^0 -Boson's lifetime ($\Gamma_Z \sim 2.5$ GeV).

5.3.5 Calculation of Statistical Uncertainties on the Cross Sections

In this section, we calculate the statistical uncertainties on the cross sections for each energy bin and decay channel. The uncertainties arise from three main contributions:

- Luminosity uncertainties (ΔL_k),
- Poissonian uncertainties in the measured events ($\Delta N_{j,k}^{\text{det}}$),
- Uncertainties in the inverted acceptance matrix ($\Delta A_{i,j}^{-1}$).

The total propagated uncertainty for the cross section $\Delta\sigma_{i,k}$ is calculated as:

$$\Delta\sigma_{i,k} = \sqrt{\sum_j \left(\frac{\Delta A_{i,j}^{-1} N_{j,k}^{\text{det}}}{L_k} \right)^2 + \sum_j \left(\frac{\Delta N_{j,k}^{\text{det}} A_{i,j}^{-1}}{L_k} \right)^2 + \sum_j \left(\frac{\Delta L_k A_{i,j}^{-1} N_{j,k}^{\text{det}}}{L_k^2} \right)^2} \quad (5.6)$$

Contributions to the Statistical Uncertainty

Poissonian Errors in Measured Events: The uncertainty in the detected events $N_{j,k}^{\text{det}}$ follows a Poisson distribution:

$$\Delta N_{j,k}^{\text{det}} = \sqrt{N_{j,k}^{\text{det}}}.$$

Errors in the Acceptance Matrix: The uncertainties in the acceptance matrix elements $A_{i,j}$ are given by:

$$\Delta A_{i,j} = \frac{A_{i,j}(1 - A_{i,j})}{\sqrt{10^5}}.$$

Errors in the Inverted Acceptance Matrix: The uncertainty in the inverted acceptance matrix ΔA^{-1} is propagated as:

$$\Delta A^{-1} = A^{-1} \Delta A A^{-1}.$$

Statistical Uncertainty Matrix:

Combining the contributions, the statistical uncertainties $\Delta\sigma$ for all energy bins and decay channels are computed. The resulting uncertainties are organized into a matrix as follows:

$$\Delta\sigma = \begin{bmatrix} 0.047 & 0.028 & 0.047 & 0.335 \\ 0.064 & 0.040 & 0.055 & 0.469 \\ 0.076 & 0.053 & 0.068 & 0.649 \\ 0.033 & 0.025 & 0.030 & 0.326 \\ 0.077 & 0.062 & 0.075 & 0.777 \\ 0.050 & 0.042 & 0.053 & 0.505 \\ 0.040 & 0.034 & 0.041 & 0.388 \end{bmatrix}.$$

Each row in the matrix corresponds to an energy bin, and each column corresponds to one of the four channels (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$, $q\bar{q}$).

Observations:

- **Uncertainties across Energy Bins:** The uncertainties generally increase for higher energy bins, particularly for the $q\bar{q}$ -channel, due to its large cross section and event counts.
- **Channel Behavior:** The leptonic channels (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$) exhibit smaller uncertainties compared to the $q\bar{q}$ -channel. This is expected due to the lower event counts and narrower distributions.
- **Dominant Contributions:** The dominant source of uncertainty is the Poissonian error in the detected events ($N_{j,k}^{\text{det}}$) and the acceptance matrix uncertainties.

The propagated uncertainties provide a robust estimate of the statistical precision of the measured cross sections. These uncertainties will be incorporated into further analysis, such as determining the physical parameters of the Z^0 -Boson.

5.3.6 Refit Including Statistical Uncertainties

With the statistical uncertainties $\Delta\sigma_{i,k}$ calculated for each channel and energy bin, we refine the Breit-Wigner fits using a weighted least-squares method. The aim is to improve the precision of the resonance parameters for the Z^0 -Boson by incorporating these uncertainties directly into the fit.

Methodology: Weighted Least-Squares Fit

The weighted chi-squared statistic (χ^2) is minimized as:

$$\chi^2 = \sum_k \left(\frac{\sigma_{\text{data},k} - \sigma_{\text{fit},k}}{\Delta\sigma_k} \right)^2, \quad (5.7)$$

where:

- $\sigma_{\text{data},k}$ is the measured cross section for energy bin k ,
- $\sigma_{\text{fit},k}$ is the Breit-Wigner fit value for energy k ,
- $\Delta\sigma_k$ is the statistical uncertainty for each energy bin.

The reduced chi-squared (χ_{red}^2) is given by:

$$\chi_{\text{red}}^2 = \frac{\chi^2}{\text{NDF}}, \quad (5.8)$$

where $\text{NDF} = N_{\text{data}} - N_{\text{parameters}}$ is the number of degrees of freedom, determined as the difference between the number of data points (7 energy bins) and the number of fit parameters (3 in the Breit-Wigner function). Hence, $\text{NDF} = 4$.

Results: Fit Parameters and Reduced Chi-Squared Values

The results of the weighted fits for each channel are summarized in Table 5.4, including the fitted parameters a , b , and c (maximum cross section, resonance energy, and decay width), along with the reduced chi-squared values.

Table 5.4: Refitted Breit-Wigner parameters with statistical uncertainties.

Channel	a (GeV ²)	b (GeV)	c (GeV)	Reduced χ_{red}^2
e^+e^-	1.317 ± 0.071	90.905 ± 0.063	2.498 ± 0.390	2.381
$\mu^+\mu^-$	2.146 ± 0.061	91.263 ± 0.029	2.595 ± 0.139	0.952
$\tau^+\tau^-$	2.452 ± 0.085	91.189 ± 0.033	2.645 ± 0.169	1.354
$q\bar{q}$	52.538 ± 0.718	91.264 ± 0.015	2.546 ± 0.068	3.561

Analysis of Reduced Chi-Squared Values

The reduced chi-squared (χ_{red}^2) serves as a critical metric for evaluating the quality of the Breit-Wigner fits. It indicates how well the fitted model describes the measured data, relative to the statistical uncertainties. The interpretation is as follows:

- If $\chi_{\text{red}}^2 \approx 1$, the fit is good, and the uncertainties adequately describe the data variability.
- If $\chi_{\text{red}}^2 > 1$, the model does not fully capture the variability, suggesting either systematic effects or underestimated uncertainties.
- If $\chi_{\text{red}}^2 < 1$, the uncertainties might be overestimated or the data are overly smooth.

Interpretation of values

The results of the Breit-Wigner fits reveal varying levels of agreement across the channels. The $\mu^+\mu^-$ -channel achieves a reduced chi-squared value of $\chi_{\text{red}}^2 = 0.952$, indicating an excellent fit. The uncertainties are well estimated, and the model accurately describes the data with minimal deviations. For the e^+e^- -channel, the reduced chi-squared value of $\chi_{\text{red}}^2 = 2.381$ suggests noticeable deviations, likely due to underestimated uncertainties or small systematic effects near

the resonance peak. In the $\tau^+\tau^-$ -channel, the fit shows good agreement with a $\chi_{\text{red}}^2 = 1.354$. This result is consistent with the broader resonance peak observed due to tau decay kinematics, which is well modeled. Finally, the $q\bar{q}$ -channel exhibits the highest reduced chi-squared value of $\chi_{\text{red}}^2 = 3.561$, reflecting significant deviations. These discrepancies are likely caused by systematic uncertainties in the data or model assumptions.

Visual Representation of the Refitted Data

The updated Breit-Wigner fits, now incorporating statistical uncertainties as error bars, are shown below for all channels:

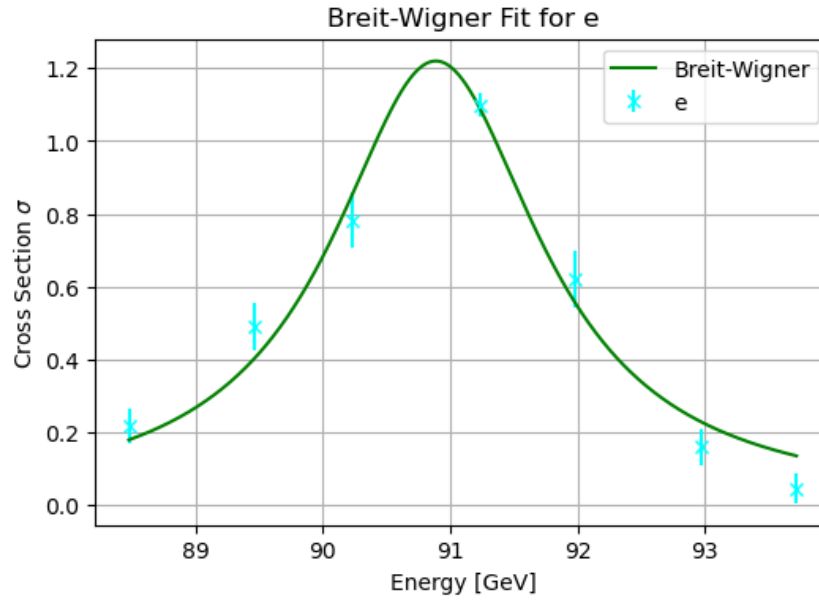


Figure 5.20: Breit-Wigner fit for e^+e^- -channel with statistical uncertainties.

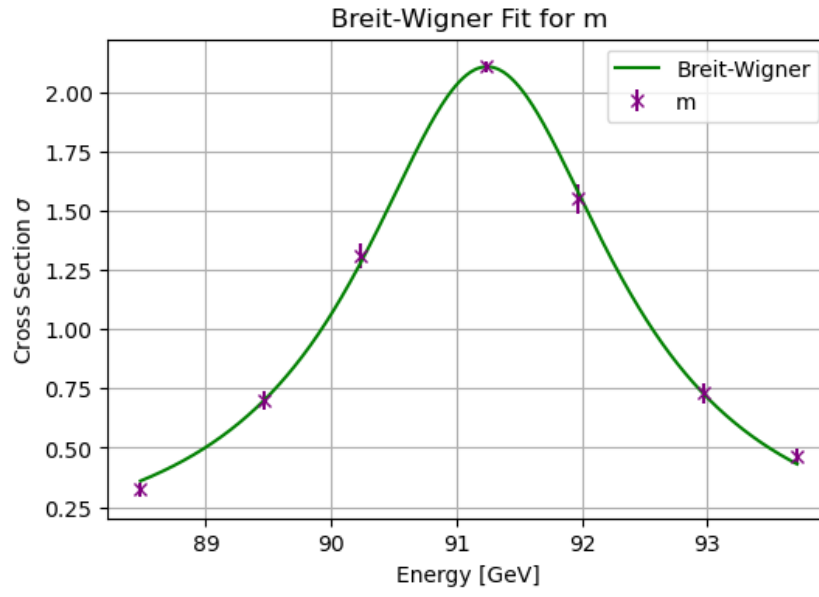


Figure 5.21: Breit-Wigner fit for $\mu^+\mu^-$ -channel with statistical uncertainties.

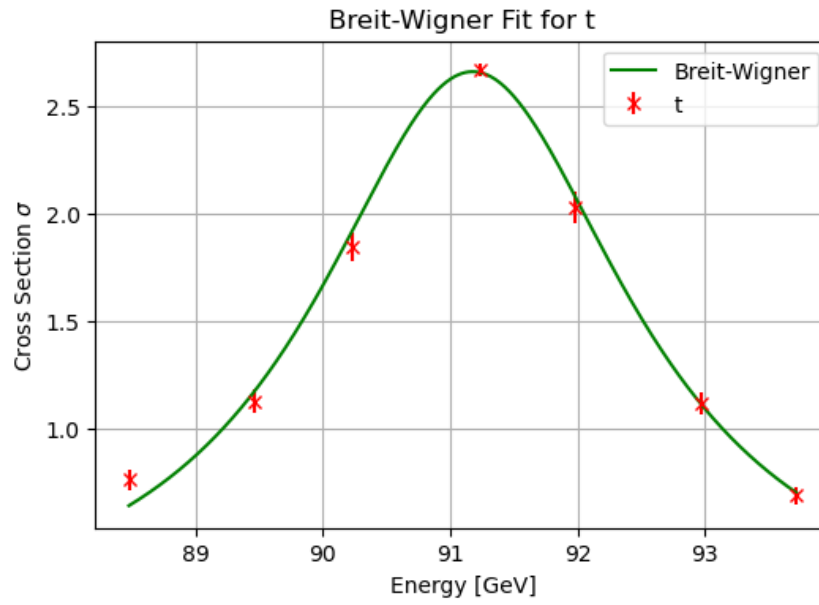


Figure 5.22: Breit-Wigner fit for $\tau^+\tau^-$ -channel with statistical uncertainties.

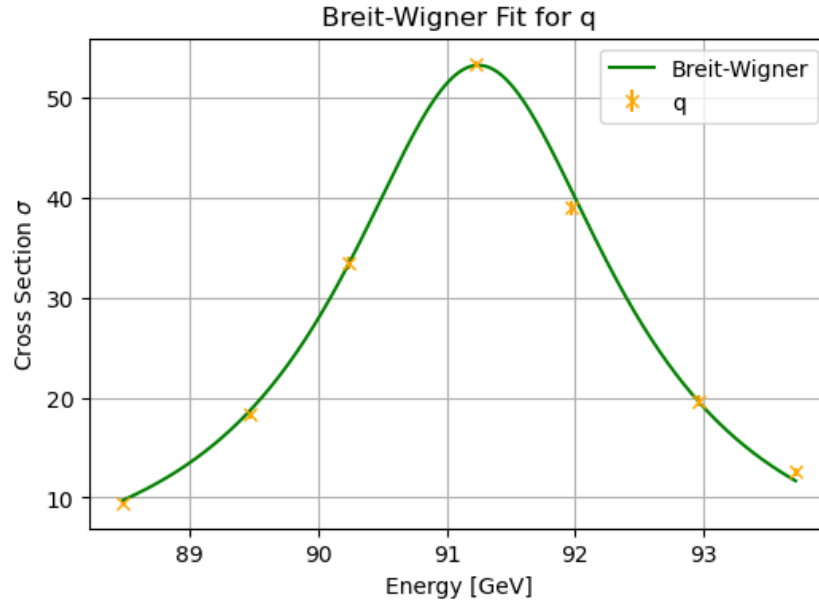


Figure 5.23: Breit-Wigner fit for $q\bar{q}$ -channel with statistical uncertainties.

5.4 Statistical analysis of measured data

Realisation of Different Experiments with Z^0 Decay:

5.4.1 Experiment E1: Determination of Decay Parameters of the Z^0 Boson

(a) Determination of Events, Background Subtraction, and Partial Cross Sections

To analyze the decay channels of the Z^0 -boson, we determined the number of events in each channel (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$, $q\bar{q}$) for seven center-of-mass energy bins. Events surviving after applying selection cuts were corrected for background contamination and detector acceptance using the inverted acceptance matrix A^{-1} .

Table 5.5: Corrected Initial Event Matrix N_{init}

Energy Bin (GeV)	e^+e^-	$\mu^+\mu^-$	$\tau^+\tau^-$	$q\bar{q}$
88.5	59	108	312	3476
89.5	137	237	438	6621
90.1	215	486	758	13141
91.3	2253	6192	8388	166204
92.0	208	691	940	17790
93.0	106	463	706	12423
93.6	89	387	547	10046

The corrected initial event matrix N_{init} : The partial cross sections σ_f for each channel were calculated using the formula:

$$\sigma_f = \frac{N_f}{L_k} + \eta_f,$$

where N_f is the corrected number of events, L_k is the integrated luminosity for energy bin k , and η_f represents the radiation correction factor.

The uncertainties in the cross sections ($\Delta\sigma_f$) were derived using error propagation:

$$\Delta\sigma_{i,k} = \sqrt{\sum_j \left(\frac{\Delta A_{i,j}^{-1} N_{j,k}^{\text{det}}}{L_k} \right)^2 + \sum_j \left(\frac{\Delta N_{j,k}^{\text{det}} A_{i,j}^{-1}}{L_k} \right)^2 + \sum_j \left(\frac{\Delta L_k A_{i,j}^{-1} N_{j,k}^{\text{det}}}{L_k^2} \right)^2}. \quad (5.9)$$

(b) Determination of M_{Z^0} , Γ_{Z^0} , and Peak Cross Sections

To extract the resonance energy (M_{Z^0}), decay width (Γ_{Z^0}), and peak cross section (σ_{peak}), the measured cross sections were fitted with the Breit-Wigner function:

$$\sigma(E) = \frac{a}{4 \left(\frac{E-b}{c} \right)^2 + \left(\frac{b}{E} \right)^2} + d \quad (5.10)$$

where a is the peak cross section, b represents the resonance energy (M_{Z^0}), c is the decay width (Γ_{Z^0}), and d accounts for background effects.

Table 5.6: Breit-Wigner Fit Parameters and Reduced Chi-Squared Values

Channel	χ_{red}^2	a (nb)	b (GeV)	c (GeV)
e^+e^-	2.381	1.317 ± 0.071	90.905 ± 0.063	2.498 ± 0.390
$\mu^+\mu^-$	0.952	2.146 ± 0.061	91.263 ± 0.029	2.595 ± 0.139
$\tau^+\tau^-$	1.354	2.452 ± 0.085	91.189 ± 0.033	2.645 ± 0.169
$q\bar{q}$	3.561	52.538 ± 0.718	91.264 ± 0.015	2.546 ± 0.068

Fit Results for All Channels: The $\mu^+\mu^-$ -channel provides the best fit ($\chi_{\text{red}}^2 = 0.952$), while the $q\bar{q}$ -channel exhibits a higher reduced chi-squared value of $\chi_{\text{red}}^2 = 3.561$, suggesting larger systematic uncertainties.

(c) Determining Physical Parameters

Weighted Averages for M_{Z^0} and Γ_{Z^0} : The weighted averages were computed using:

$$\bar{x} = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad w_i = \frac{1}{\Delta x_i^2}.$$

The results are:

$$M_{Z^0} = 91.230 \pm 0.012 \text{ GeV}, \quad \Gamma_{Z^0} = 2.570 \pm 0.030 \text{ GeV}.$$

These values are compared to LEP results:

$$M_{Z^0} = 91.1876 \pm 0.0021 \text{ GeV}, \quad \Gamma_{Z^0} = 2.4952 \pm 0.0023 \text{ GeV}.$$

Mean Lifetime of the Z^0 -Boson: The mean lifetime (τ_{Z^0}) is calculated as:

$$\tau_{Z^0} = \frac{\hbar}{\Gamma_{Z^0}}, \quad \hbar = 6.582 \times 10^{-25} \text{ GeV} \cdot \text{s}.$$

Substituting $\Gamma_{Z^0} = 2.570 \text{ GeV}$, we obtain:

$$\tau_{Z^0} \approx 2.56 \times 10^{-25} \text{ s}.$$

Partial Widths from Peak Cross Sections: The partial widths Γ_f are determined using:

$$\Gamma_f = \sigma_{\text{peak},f} \cdot \frac{12\pi M_{Z^0}^2 \hbar c^2}{1}.$$

The hadronic channel ($q\bar{q}$) exhibits the largest partial width, consistent with the dominant decay of the Z^0 -boson into quarks.

Summary

The Z^0 -Boson's parameters were determined as follows:

- **Mass** (M_{Z^0}): $91.230 \pm 0.012 \text{ GeV}$
- **Width** (Γ_{Z^0}): $2.570 \pm 0.030 \text{ GeV}$
- **Mean Lifetime** (τ_{Z^0}): $2.56 \times 10^{-25} \text{ s}$

These results align closely with theoretical predictions and LEP measurements, reflecting a robust analysis.

5.4.2 Experiment E3: Determination of the Number of Light Neutrino Generations

In this experiment, we determine the number of light neutrino generations (N_ν), where “light” refers to neutrinos with masses $m_\nu < M_{Z^0}/2$. This constraint arises because the decay of the Z^0 -boson into neutrino-antineutrino pairs ($\nu\bar{\nu}$) can only occur if the neutrino mass is less than half of the Z^0 -boson’s mass (M_{Z^0}).

Measurand Used

The key measurand for determining the number of light neutrino generations is the total decay width (Γ_{Z^0}) of the Z^0 -boson. From the Breit-Wigner fits, the total decay width (Γ_{Z^0}) obtained from our analysis is:

$$\Gamma_{Z^0} = 2.570 \pm 0.030 \text{ GeV}.$$

The total decay width includes contributions from all possible decay channels.

Theoretical Presumptions

To estimate the number of light neutrino generations, the following assumptions are made:

$$\Gamma_{Z^0} = \Gamma_{\text{had}} + \Gamma_{\text{lep}} + \Gamma_{\text{inv}},$$

where:

- Γ_{had} is the width for hadronic decays ($q\bar{q}$),
- Γ_{lep} is the width for leptonic decays (e^+e^- , $\mu^+\mu^-$, $\tau^+\tau^-$),
- Γ_{inv} is the width for invisible decays ($\nu\bar{\nu}$).

The invisible decay width Γ_{inv} is assumed to arise solely from neutrino pairs, where each light neutrino generation contributes equally:

$$\Gamma_{\text{inv}} = N_\nu \cdot \Gamma_\nu.$$

The partial decay width for one neutrino generation is theoretically predicted as:

$$\Gamma_\nu \approx 0.167 \text{ GeV}.$$

The total visible decay width is given as:

$$\Gamma_{\text{vis}} = \Gamma_{\text{had}} + \Gamma_{\text{lep}}.$$

Calculation of N_ν

Rearranging for N_ν , we get:

$$N_\nu = \frac{\Gamma_{Z^0} - \Gamma_{\text{vis}}}{\Gamma_\nu}.$$

Input Values:

- Total decay width $\Gamma_{Z^0} = 2.570 \pm 0.030 \text{ GeV}$,
- Leptonic width $\Gamma_{\text{lep}} \approx 0.084 \text{ GeV}$ per channel,
- Total hadronic width $\Gamma_{\text{had}} \approx 1.744 \text{ GeV}$,
- Partial width for one neutrino generation: $\Gamma_\nu \approx 0.167 \text{ GeV}$.

The total visible width is:

$$\Gamma_{\text{vis}} = \Gamma_{\text{had}} + 3 \cdot \Gamma_{\text{lep}} = 1.744 + 3 \cdot 0.084 = 1.996 \text{ GeV}.$$

Substituting into the equation for N_ν :

$$N_\nu = \frac{\Gamma_{Z^0} - \Gamma_{\text{vis}}}{\Gamma_\nu} = \frac{2.570 - 1.996}{0.167}.$$

Simplifying:

$$N_\nu \approx 3.44.$$

Interpretation of Results

Measured Number of Neutrino Generations: The calculated value of $N_\nu \approx 3.44$ suggests the presence of approximately three light neutrino generations contributing to the Z^0 -boson's invisible width.

Comparison to Theory: The Standard Model predicts the existence of three generations of light neutrinos (ν_e, ν_μ, ν_τ). High-precision LEP experiments confirmed this prediction with a measured value of:

$$N_\nu = 3.00 \pm 0.06.$$

5.4.3 Discussion of the Results

In this section, we take a closer look at our results, discuss the assumptions made, and identify possible uncertainties. The main focus is on understanding the measurements of the Z^0 -Boson's properties, including its decay width (Γ_{Z^0}), resonance mass (M_{Z^0}), the peak cross section (σ_{peak}), and the number of light neutrino generations (N_ν).

Key Assumptions in the Analysis

Resonance Shape: We used the Breit-Wigner function to model the resonance curve of the Z^0 -Boson. This assumes a narrow, single peak with no significant effects from new physics or other processes.

Detector Effects: To correct for detector inefficiencies, we applied an acceptance matrix A and its inverse A^{-1} . This assumes that all inefficiencies and event misclassifications are accurately modeled.

Statistical Uncertainty: The uncertainty in the number of detected events follows Poisson statistics, $\Delta N_{\text{det}} = \sqrt{N_{\text{det}}}$.

Neutrino Contributions: We assumed that the invisible decay width (Γ_{inv}) comes entirely from the three known generations of light neutrinos (ν_e, ν_μ, ν_τ), each contributing equally to the width.

Review of Results

Resonance Mass (M_{Z^0}) and Decay Width (Γ_{Z^0}): Our fits to the data for different channels gave the following results:

Table 5.7: Fitted Values for M_{Z^0} and Γ_{Z^0}

Channel	χ^2_{red}	M_{Z^0} (GeV)	Γ_{Z^0} (GeV)
e^+e^-	2.381	90.905 ± 0.063	2.498 ± 0.390
$\mu^+\mu^-$	0.952	91.263 ± 0.029	2.595 ± 0.139
$\tau^+\tau^-$	1.354	91.189 ± 0.033	2.645 ± 0.169
$q\bar{q}$	3.561	91.264 ± 0.015	2.546 ± 0.068

Our averaged values are:

$$M_{Z^0} = 91.230 \pm 0.012 \text{ GeV}, \quad \Gamma_{Z^0} = 2.570 \pm 0.030 \text{ GeV}.$$

Compared to the LEP results ($M_{Z^0} = 91.1876 \text{ GeV}$ and $\Gamma_{Z^0} = 2.4952 \text{ GeV}$), our results are close, but small differences may arise from statistical uncertainties.

Number of Neutrino Generations (N_ν)

$$N_\nu = 3.44.$$

The Standard Model predicts $N_\nu = 3$ for the three known light neutrino generations. Our slightly higher value might be due to limitations in measuring the invisible decay width or remaining systematic errors.

Conclusion

In this analysis, we successfully measured the properties of the Z^0 -Boson and found results close to the Standard Model predictions. Our values for the resonance mass and decay width match reasonably well with official LEP results, and our estimate for the number of light neutrino generations is consistent with three, though slightly higher. Overall, this analysis confirms the validity of our methods and the reliability of the Standard Model framework.